The evolution terms in colour-singlet form

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1 Summary

What is done. Every evolution term, i.e. every two-ρ term multiplying $\log \frac{\vee}{\Lambda}$, is written in the colour-singlet form of your Appendices E and F:

- the projectile is taken in a colour-singlet state, $\langle \rho^a(x)\rho^b(y)\rangle = \delta^{ab}\,\langle \rho^f(x)\rho^f(y)\rangle/(N_c^2-1)$;
- the colour factor is then a closed network of adjoint Wilson lines;
- it is written with traces $\mathrm{Tr}\{\dots\}$ of U, $U^\dagger$ and the adjoint generators T^e (Sec. 2).

The traces are produced by a program that converts each term automatically. For each of the 489 terms of your lists and each of the 7932 terms of the JIMWLK and BFKL targets, the trace form was compared with the direct contraction of the colour network, using random general SU(3) Wilson lines. They agree to 2.2×10^{-15}.

Results.

1. **LO** is your Eq. (LOx), Eq. (2.2).
2. **JIMWLK⊗LO** (the total coefficient of $\log \frac{\vee}{\Lambda}$) is Eq. (3.1): 44 traces in 8 kernel structures. It equals your App. F with the corrected M (M2 of `Final_evolution_check.pdf`) for a general U: relative difference 1.3×10^{-15}.
3. **BFKL⊗LO** (the coefficient of $\log \vee$) is Eq. (3.2). It is your App. E times $(-N_c)$, line by line (8 terms, relative difference 4.2×10^{-16}).
4. **Leading-log result** in singlet form: Eq. (3.3).
5. **Row by row**, Sec. 4: every row of Groups I, II, III in singlet form, with the index placement valid for a general adjoint U.
   - The rows come from the leading-log wave function (the region-T reference rows of the earlier notes).
   - Their sum is JIMWLK⊗LO for a general U (relative difference 1.8×10^{-15} for $U=U^T$, 1.4×10^{-15} for a general U).
6. **Your lists** (`Evolution_terms.tex` Secs. 1, 2 and the `4rho.tex` lists) in singlet form, exactly as written: App. A.

One thing to note: $U^{bc} = U^{cb}$. Your lists use the convention $U^{bc} = U^{cb}$, i.e. $U^\dagger = U$.

- **With this convention**, your rows and the general-U rows of Sec. 4 agree:
  - row by row for Groups II and III (except Group II Row IV);
  - as a whole for Group I together with Group II Row IV (Table 2);
  - kernel structure by kernel structure in the total (Table 1).
- **For a general adjoint** U ($U^\dagger \neq U$, as for a real target) your lists, read with the index order as written, do *not* add up to JIMWLK⊗LO: 11 of the 13 kernel structures differ, by 3.8 (relative, at random points) in total.
- No simple reading rule repairs this (we tried "every Wilson line acts on its charge"). The information on which index comes first is not in the lists.
- The physical (general U) row-by-row forms are those of Sec. 4. Your App. E and App. F (corrected) are already written for a general U and are correct.

This matters only for the row-by-row presentation. The sum, Eqs. (3.1), (3.2), (3.3), is correct as it stands.

2 Projection, notation and the trace rules

Normalization. Every row below is written as a bracket $\{\dots\}$ with

$$\frac{d^3N}{d^3k} = \mathcal{N} \log\frac{\vee}{\Lambda} \frac{1}{N_c^2-1} \int_{x,y,z,w,w'} e^{-ik\cdot(w'-w)} \langle \rho^f(x)\rho^f(y)\rangle \{\dots\}, \qquad \mathcal{N} = \frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+} = \frac{\alpha_s^2}{8\pi^6 k^+}. \tag{2.1}$$

So $\mathcal{N}\log\frac{\vee}{\Lambda}/(N_c^2-1) = P/k^+$ with your $P = \frac{\alpha_s^2\delta Y}{8\pi^6(N_c^2-1)}$ and $\delta Y = \log\frac{\vee}{\Lambda}$. The brackets are therefore directly those of your App. E and App. F (there the soft point is called t; here it is z).

- x, y are the two charge points. $\langle\rho^f(x)\rho^f(y)\rangle$ is symmetric and both are integrated, so a term and its $x \leftrightarrow y$ image are equal. Each term is written with x attached to w where possible.
- Kernels: $\mathcal{K}(v) = v/v^2$, $K(a,b,z) \equiv \mathcal{K}(a-z)\cdot\mathcal{K}(b-z)$, $K(a,a,z) = \frac{1}{(a-z)^2}$.

The projection. $\rho^a(x)\rho^b(y) \to \delta^{ab}\langle\rho^f(x)\rho^f(y)\rangle/(N_c^2-1)$. After it, all colour indices are contracted among Wilson lines and structure constants.

Trace rules (exact for any SU(N_c) adjoint U):

- $(T^e)_{bc} = -if^{ebc}$, so $f^{abc} = i(T^a)_{bc}$.
- U is real and orthogonal: $U^T = U^\dagger = U^{-1}$. A chain $\dots U^{ab}(p)\dots$ read from b to a gives $U^\dagger(p)$.
- Two structure constants connected by a chain C of Wilson lines: $\sum_d C^{ad}T^d = C^\dagger T^a C$. Hence every two-$f$ network is $\mathrm{Tr}\{T^eAT^eB\}$ ("theta") or $\mathrm{Tr}\{AT^e\}\mathrm{Tr}\{BT^e\}$ ("dumbbell").
- $U(p)U^\dagger(p) = 1$, $\mathrm{Tr}\,1 = N_c^2-1$, $T^eT^e = N_c$, $\mathrm{Tr}\,T^e = 0$, $\mathrm{Tr}\{T^eAT^eA^\dagger\} = N_c\mathrm{Tr}A$.
- $\mathrm{Tr}\{U^\dagger(p)U(q)\} = \mathrm{Tr}\{U^\dagger(q)U(p)\}$ (real U).

Dipole combinations are collected into your LO structure,

$$\mathrm{Tr}\{(U^\dagger(a)-U^\dagger(w))(U(b)-U(w'))\} = \mathrm{Tr}\{U^\dagger(a)U(b)\} - \mathrm{Tr}\{U^\dagger(a)U(w')\} - \mathrm{Tr}\{U^\dagger(w)U(b)\} + \mathrm{Tr}\{U^\dagger(w)U(w')\},$$

with $\mathrm{Tr}\{U^\dagger(z)U(z)\} = N_c^2-1$.

LO. In the units of LO, $\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2k^+}\frac{1}{N_c^2-1}\int_{x,y,w,w'} e^{-ik\cdot(w'-w)}\langle\rho^f(x)\rho^f(y)\rangle\{\dots\}$:

$$\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\Big[\mathrm{Tr}\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\}\Big] \tag{2.2}$$

This is your Eq. (LOx).

3 The totals

3.1 JIMWLK⊗LO: the coefficient of $\log\frac{\vee}{\Lambda}$

Bracket of Eq. (2.1), for a general U:

$$
\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(w,w',z)\Big[4N_c\mathrm{Tr}\{U^\dagger(w)U(w')\}\\
&\qquad-2\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(w')U(z)\}-2\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(z)U(w)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w',z)\Big[-4N_c\mathrm{Tr}\{U^\dagger(x)U(w')\}\\
&\qquad+2\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(x)U(z)\}+2\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(w')\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,y,z)\Big[4N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad-2\mathrm{Tr}\{T^eU^\dagger(x)U(y)T^eU^\dagger(y)U(z)\}-2\mathrm{Tr}\{T^eU^\dagger(x)U(y)T^eU^\dagger(z)U(x)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w,z)\Big[-4N_c\mathrm{Tr}\{U^\dagger(w)U(y)\}\\
&\qquad+2\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(y)U(z)\}+2\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(z)U(w)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(w',w',z)\Big[-2N_c\mathrm{Tr}\{U^\dagger(w)U(w')\}+2N_c\mathrm{Tr}\{U^\dagger(x)U(w')\}\\
&\qquad-\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(w')U(z)\}-\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(w')\}\\
&\qquad+\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(w')U(z)\}+\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(z)U(w')\}\\
&\qquad+\mathrm{Tr}\{T^eU^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(w')U(z)\}\\
&\qquad+\mathrm{Tr}\{T^eU^\dagger(w)U(w')\}\,\mathrm{Tr}\{T^eU^\dagger(w')U(z)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(w,w,z)\Big[-2N_c\mathrm{Tr}\{U^\dagger(w)U(w')\}+2N_c\mathrm{Tr}\{U^\dagger(w)U(y)\}\\
&\qquad+\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(w)U(z)\}+\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(z)U(w)\}\\
&\qquad-\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(w)U(z)\}-\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(z)U(w)\}\\
&\qquad-\mathrm{Tr}\{T^eU^\dagger(w)U(w')\}\,\mathrm{Tr}\{T^eU^\dagger(w)U(z)\}\\
&\qquad+\mathrm{Tr}\{T^eU^\dagger(w)U(y)\}\,\mathrm{Tr}\{T^eU^\dagger(w)U(z)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,x,z)\Big[2N_c\mathrm{Tr}\{U^\dagger(x)U(w')\}-2N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad-\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(x)U(z)\}-\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(x)\}\\
&\qquad+\mathrm{Tr}\{T^eU^\dagger(x)U(y)T^eU^\dagger(x)U(z)\}+\mathrm{Tr}\{T^eU^\dagger(x)U(y)T^eU^\dagger(z)U(x)\}\\
&\qquad-\mathrm{Tr}\{T^eU^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(x)U(z)\}\\
&\qquad-\mathrm{Tr}\{T^eU^\dagger(x)U(y)\}\,\mathrm{Tr}\{T^eU^\dagger(x)U(z)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,y,z)\Big[2N_c\mathrm{Tr}\{U^\dagger(w)U(y)\}-2N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad-\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(y)U(z)\}-\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(z)U(y)\}\\
&\qquad+\mathrm{Tr}\{T^eU^\dagger(x)U(y)T^eU^\dagger(y)U(z)\}+\mathrm{Tr}\{T^eU^\dagger(x)U(y)T^eU^\dagger(z)U(y)\}\\
&\qquad-\mathrm{Tr}\{T^eU^\dagger(w)U(y)\}\,\mathrm{Tr}\{T^eU^\dagger(y)U(z)\}\\
&\qquad+\mathrm{Tr}\{T^eU^\dagger(x)U(y)\}\,\mathrm{Tr}\{T^eU^\dagger(y)U(z)\}\Big]
\end{aligned}
\tag{3.1}
$$

- This is your App. F with M at the points the charges act on (correction M2). For example, the $K(x,y,z)$ line is $2\,\mathrm{Tr}\{T^eU^\dagger(x)U(y)T^e(U^\dagger(y)-U^\dagger(z))(U(x)-U(z))\}$, i.e. $dN^{(1)}+dN^{(2)}$, since $dN^{(1)}=dN^{(2)}$.
- The two-trace terms are $dN^{(5)}$.
- Numerically, Eq. (3.1) equals App. F (corrected): relative difference 1.3×10^{-15}.

3.2 BFKL⊗LO: the coefficient of $\log\vee$

Bracket of Eq. (2.1) (with $\log\frac{\vee}{\Lambda}\to\log\vee$), for a general U:

$$
\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,y,z)\Big[2N_c\mathrm{Tr}\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\}\Big]
$$

$$
\begin{aligned}
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,x,z)\Big[-N_c\mathrm{Tr}\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,y,z)\Big[-N_c\mathrm{Tr}\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\}\Big]\\
&+K(w,w',z)\,K(x,y,z)\Big[2N_c\mathrm{Tr}\{(U^\dagger(z)-U^\dagger(w))(U(z)-U(w'))\}\Big]\\
&+\mathcal{K}(x-w')\cdot\mathcal{K}(w-z)\,K(x,y,z)\Big[2N_c\mathrm{Tr}\{(U^\dagger(z)-U^\dagger(w))(U(x)-U(w'))\}\Big]\\
&+\mathcal{K}(x-w')\cdot\mathcal{K}(w-z)\,K(y,y,z)\Big[-N_c\mathrm{Tr}\{(U^\dagger(z)-U^\dagger(w))(U(x)-U(w'))\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,K(x,y,z)\Big[2N_c\mathrm{Tr}\{(U^\dagger(x)-U^\dagger(w))(U(z)-U(w'))\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,K(y,y,z)\Big[-N_c\mathrm{Tr}\{(U^\dagger(x)-U^\dagger(w))(U(z)-U(w'))\}\Big]
\end{aligned}
\tag{3.2}
$$

Line by line this is your Eq. (LOBFKL) times $-N_c$ (in your normalization the prefactor is $-\frac{\alpha_s^2 N_c \delta Y}{8\pi^6(N_c^2-1)}$), with $K_D(x,y,z)=K(x,x,z)-2K(x,y,z)+K(y,y,z)$. The pairs ($x$-line, y-line) of your App. E appear here with $x\leftrightarrow y$ where needed: $\mathcal{K}(x-w')\cdot\mathcal{K}(w-z)\,[\dots]$ is your $\mathcal{K}(z-w)\cdot\mathcal{K}(y-w')$ line. Numerically: relative difference 4.2×10^{-16}.

3.3 The leading-log result

With your rule A, $\log\frac{\Lambda}{k^+}=-\log\frac{\vee}{\Lambda}+\log\frac{\vee}{k^+}$:

$$
\boxed{\frac{d^3N_{\rm LL}}{d^3k}=\frac{\mathcal{N}}{N_c^2-1}\int_{x,y,z,w,w'}e^{-ik\cdot(w'-w)}\langle\rho^f(x)\rho^f(y)\rangle\Big[\log\frac{\vee}{\Lambda}\,\{\text{Eq. (3.1)}\}+\log\frac{\vee}{k^+}\Big(\{\text{Eq. (3.2)}\}-\{\text{Eq. (3.1)}\}\Big)\Big]}
\tag{3.3}
$$

- Equivalently, $\log\frac{k^+}{\Lambda}\{\text{Eq. (3.1)}\}+\log\frac{\vee}{k^+}\{\text{Eq. (3.2)}\}$.
- At mid rapidity, $k^+=\sqrt{\vee\Lambda}$, it is $\frac{1}{2}\log\frac{\vee}{\Lambda}\big(\{\text{Eq. (3.1)}\}+\{\text{Eq. (3.2)}\}\big)$.
- Three and four ρ's: their symmetrized parts cancel identically (earlier notes), so they give nothing in any projection. Their commutator parts are two-ρ terms and are included in Eqs. (3.1) and (3.2) and in the rows below.

3.4 Kernel structure by kernel structure

Your lists, summed and compared with Eq. (3.1) for each kernel structure. Each entry is the difference relative to the sum of the moduli of the single terms, at random points.

Table 1: Sum of your lists minus JIMWLK⊗LO, singlet projection.

kernel structure	in Eq. (3.1)	$U=U^T$ (your convention)	general U
$\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(w,w',z)$	yes	1.3×10^{-16}	6.2×10^{-1}
$\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w',z)$	yes	2.2×10^{-16}	3.6×10^{-1}
$\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w,z)$	no	4.7×10^{-17}	5.1×10^{-1}
$\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,y,z)$	yes	2.1×10^{-16}	1.8×10^{-1}
$\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w',z)$	no	1.6×10^{-16}	3.1×10^{-1}
$\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w,z)$	yes	1.8×10^{-16}	2.5×10^{-1}
$\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(w',w',z)$	yes	2.0×10^{-16}	2.5×10^{-16}
$\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(w,w,z)$	yes	1.6×10^{-16}	1.3×10^{-16}
$\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,x,z)$	yes	1.0×10^{-16}	3.6×10^{-1}
$\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,y,z)$	yes	5.8×10^{-17}	6.0×10^{-1}
$\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(x,y,z)$	no	9.1×10^{-17}	8.8×10^{-1}
$\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w',z)$	no	1.0×10^{-16}	5.1×10^{-1}
$\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w,z)$	no	6.8×10^{-17}	3.6×10^{-1}

The five structures marked "no" are not in Eq. (3.1). Their sum over your rows vanishes for $U=U^T$.

4 Row by row, for a general U

Where these rows come from. The region-T rows of the earlier notes:

- they are built directly from the leading-log light-cone wave function, with the operator order of the charges kept;
- every Wilson line is placed as it acts, i.e. correct for a general U;

- they are written with the pieces $\mathfrak{V}$ of `NLO_gluon_midrapidity.pdf`, Sec. 6.

For each row:

- the three- and four-ρ products are decomposed into fully symmetric parts plus commutators, with $[\rho^a(u), \rho^b(v)] = if^{abc}\rho^c(u)\delta(u-v)$;
- the two-ρ part is kept;
- it is projected and written as traces.

Coefficients such as $\frac{1}{6}$, $\frac{1}{3}$ come from this symmetrization. The split between rows depends on it; the sum does not. Group I contains the c.c.; Group III contains the c.c.

Table 2: Your rows (Secs. 1, 2 of `Evolution_terms.tex`, with the two-ρ part of each line of Eq. (4rho) attributed to its row) against the rows below, singlet projection, relative difference.

row below	your row	$U = U^T$	general U
Group I, Row I(a)+I(b)	two-ρ part of lines 1, 2 of (4rho)	4.5×10^{-1}	4.1×10^{-1}
Group I, Row II(a)+II(b)	genuine + from three ρ's	8.2×10^{-1}	9.1×10^{-1}
Group I, Row III	from three ρ's (N_1, N_2) + line 4 of (4rho)	1.8×10^{0}	2.0×10^{0}
Group I, Row IV	from three ρ's (N_1, N_2) + line 3 of (4rho)	1.1×10^{0}	1.4×10^{0}
Group I, Row V	genuine + from three ρ's	1.1×10^{0}	1.5×10^{0}
Group I, Row I′	— (inside your Group I)	1.0×10^{0}	1.0×10^{0}
Group II, Row I	genuine + from three ρ's + line 5 of (4rho)	3.5×10^{-16}	1.7×10^{0}
Group II, Row II	line 6 of (4rho)	6.2×10^{-15}	1.1×10^{0}
Group II, Row III	line 7 of (4rho)	0	3.1×10^{-1}
Group II, Row IV	genuine	1.2×10^{0}	1.2×10^{0}
Group III, Row I	from three ρ's (N_1, N_2) + line 8 of (4rho)	6.4×10^{-15}	8.4×10^{-1}
Group III, Row II	from three ρ's (N_1, N_2) + line 9 of (4rho)	1.5×10^{-14}	1.1×10^{0}
Group III, Row III	genuine + from three ρ's	3.3×10^{-16}	6.1×10^{-1}
Group III, Row IV	line 10 of (4rho)	8.0×10^{-15}	1.4×10^{0}
Group III, Row V	from three ρ's	5.6×10^{-16}	8.5×10^{-15}
Group III, Row VI	from three ρ's	1.9×10^{-16}	6.2×10^{-1}
Group I (all, with I′) + Group II Row IV	the same, summed	7.1×10^{-16}	3.4×10^{-1}
all rows	all your lists	4.5×10^{-15}	8.6×10^{-1}

- In Group I the split differs: your Group I rows and Group II Row IV (whose genuine terms carry the dimensionally regulated poles written back as z integrals) distribute the terms differently. The reference also has a separate piece I′ (the measured gluon emits and reabsorbs the soft gluon), which in your lists is inside the Group I rows.
- Together they agree.
- The two-ρ parts of the lines of Eq. (4rho), attributed to their rows, equal your "From 4ρ's" list plus the `4rho.tex` lists: 3.0×10^{-12} for $U = U^T$.

4.1 Group I (with c.c.)

Group I, Row I(a).

$$
\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,y,z)\Big[-\frac{N_c}{2}\mathrm{Tr}\{U^\dagger(w)U(y)\}-\frac{N_c}{2}\mathrm{Tr}\{U^\dagger(x)U(w')\}\\
&\qquad+N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,x,z)\Big[\frac{N_c}{6}\mathrm{Tr}\{U^\dagger(w)U(y)\}+\frac{N_c}{6}\mathrm{Tr}\{U^\dagger(x)U(w')\}\\
&\qquad-\frac{N_c}{3}\mathrm{Tr}\{U^\dagger(x)U(y)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,y,z)\Big[\frac{N_c}{6}\mathrm{Tr}\{U^\dagger(w)U(y)\}+\frac{N_c}{6}\mathrm{Tr}\{U^\dagger(x)U(w')\}\\
&\qquad-\frac{N_c}{3}\mathrm{Tr}\{U^\dagger(x)U(y)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(x,y,z)\Big[-\frac{2N_c\left(N_c^2-1\right)}{3}+\frac{N_c}{3}\mathrm{Tr}\{U^\dagger(x)U(w')\}+\frac{N_c}{3}\mathrm{Tr}\{U^\dagger(x)U(w)\}\Big]
\end{aligned}
\tag{4.1}
$$

Group I, Row I(b).

$$
\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,y,z)\Big[\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(x)U(y)\}\\
&\qquad-\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(y)U(x)\}+\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(y)U(x)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,x,z)\Big[-\frac{N_c}{3}\mathrm{Tr}\{U^\dagger(w)U(w')\}+\frac{N_c}{6}\mathrm{Tr}\{U^\dagger(w)U(y)\}\\
&\qquad+\frac{N_c}{6}\mathrm{Tr}\{U^\dagger(x)U(w')\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,y,z)\Big[-\frac{N_c}{3}\mathrm{Tr}\{U^\dagger(w)U(w')\}+\frac{N_c}{6}\mathrm{Tr}\{U^\dagger(w)U(y)\}\\
&\qquad+\frac{N_c}{6}\mathrm{Tr}\{U^\dagger(x)U(w')\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(x,y,z)\Big[-\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(x)U(y)\}\\
&\qquad-\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(y)U(x)\}+\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(x)U(y)\}\\
&\qquad+\frac{1}{3}\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(y)U(x)\}-\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w)U(x)T^eU^\dagger(x)U(y)\}\\
&\qquad-\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(x)T^eU^\dagger(y)U(x)\}\\
&\qquad-\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(x)U(y)\}\\
&\qquad-\frac{1}{3}\mathrm{Tr}\{T^eU^\dagger(w)U(w')\}\,\mathrm{Tr}\{T^eU^\dagger(x)U(y)\}\\
&\qquad+\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(x)U(y)\}\Big]
\end{aligned}
\tag{4.2}
$$

Group I, Row I′.

$$
\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w',z)\Big[-N_c\mathrm{Tr}\{U^\dagger(x)U(w')\}\\
&\qquad+\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(w')U(x)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w,z)\Big[\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(x)U(w)\}\\
&\qquad-\mathrm{Tr}\{T^eU^\dagger(w)U(x)T^eU^\dagger(y)U(w)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w',z)\Big[-\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(y)U(w')\}\\
&\qquad+\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(w')U(y)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w,z)\Big[-N_c\mathrm{Tr}\{U^\dagger(w)U(y)\}\\
&\qquad+\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(y)U(w)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(w',w',z)\Big[-3\mathrm{Tr}\{U^\dagger(w)U(w')\}+3\mathrm{Tr}\{U^\dagger(x)U(w')\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(w,w,z)\Big[-3\mathrm{Tr}\{U^\dagger(w)U(w')\}+3\mathrm{Tr}\{U^\dagger(w)U(y)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w',z)\Big[\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(y)U(w')\}\\
&\qquad-\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(w')U(y)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w,z)\Big[-\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(y)U(w)\}\\
&\qquad+\mathrm{Tr}\{T^eU^\dagger(w)U(x)T^eU^\dagger(y)U(w)\}\Big]
\end{aligned}
\tag{4.3}
$$

Group I, Row II(a).

$$
\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w',z)\Big[N_c\mathrm{Tr}\{U^\dagger(w)U(w')\}-N_c\mathrm{Tr}\{U^\dagger(x)U(w')\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w,z)\Big[N_c\mathrm{Tr}\{U^\dagger(w)U(w')\}-N_c\mathrm{Tr}\{U^\dagger(w)U(y)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,y,z)\Big[-N_c\mathrm{Tr}\{U^\dagger(w)U(w')\}+\frac{N_c}{2}\mathrm{Tr}\{U^\dagger(w)U(y)\}
\end{aligned}
$$

$$
\begin{aligned}
&\quad + \frac{N_c}{2}\mathrm{Tr}\{U^\dagger(x)U(w')\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w',z)\Big[N_c\mathrm{Tr}\{U^\dagger(w)U(w')\} - N_c\mathrm{Tr}\{U^\dagger(x)U(w')\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w,z)\Big[N_c\mathrm{Tr}\{U^\dagger(w)U(w')\} - N_c\mathrm{Tr}\{U^\dagger(w)U(y)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(w',w',z)\Big[-3\mathrm{Tr}\{U^\dagger(w)U(w')\} + 3\mathrm{Tr}\{U^\dagger(x)U(w')\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(w,w,z)\Big[-3\mathrm{Tr}\{U^\dagger(w)U(w')\} + 3\mathrm{Tr}\{U^\dagger(w)U(y)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,x,z)\Big[-\frac{N_c}{3}\mathrm{Tr}\{U^\dagger(w)U(w')\} + \frac{N_c}{6}\mathrm{Tr}\{U^\dagger(w)U(y)\} \\
&\quad + \frac{N_c}{6}\mathrm{Tr}\{U^\dagger(x)U(w')\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,y,z)\Big[-\frac{N_c}{3}\mathrm{Tr}\{U^\dagger(w)U(w')\} + \frac{N_c}{6}\mathrm{Tr}\{U^\dagger(w)U(y)\} \\
&\quad + \frac{N_c}{6}\mathrm{Tr}\{U^\dagger(x)U(w')\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(x,y,z)\Big[\frac{4N_c}{3}\mathrm{Tr}\{U^\dagger(w)U(w')\} - \frac{2N_c}{3}\mathrm{Tr}\{U^\dagger(x)U(w')\} \\
&\quad - \frac{2N_c}{3}\mathrm{Tr}\{U^\dagger(x)U(w)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w',z)\Big[-N_c\mathrm{Tr}\{U^\dagger(w)U(w')\} + N_c\mathrm{Tr}\{U^\dagger(x)U(w')\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w,z)\Big[-N_c\mathrm{Tr}\{U^\dagger(w)U(w')\} + N_c\mathrm{Tr}\{U^\dagger(x)U(w)\}\Big]
\end{aligned}
\tag{4.4}
$$

Group I, Row II(b).

$$
\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,y,z)\Big[N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad - \frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(x)U(y)\} - \frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(y)U(x)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,x,z)\Big[\frac{N_c}{6}\mathrm{Tr}\{U^\dagger(w)U(y)\} + \frac{N_c}{6}\mathrm{Tr}\{U^\dagger(x)U(w')\} \\
&\quad - \frac{N_c}{3}\mathrm{Tr}\{U^\dagger(x)U(y)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,y,z)\Big[\frac{N_c}{6}\mathrm{Tr}\{U^\dagger(w)U(y)\} + \frac{N_c}{6}\mathrm{Tr}\{U^\dagger(x)U(w')\} \\
&\quad - \frac{N_c}{3}\mathrm{Tr}\{U^\dagger(x)U(y)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(x,y,z)\Big[-\frac{2N_c}{3}\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad + \frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(x)U(y)\} + \frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(y)U(x)\} \\
&\quad + \frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w)U(x)T^eU^\dagger(x)U(y)\} - \frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(x)T^eU^\dagger(y)U(x)\} \\
&\quad + \frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(x)U(y)\} \\
&\quad + \frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(x)U(y)\}\Big]
\end{aligned}
\tag{4.5}
$$

Group I, Row III.

$$
\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,y,z)\Big[\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(x)U(z)\} \\
&\quad + \frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(y)\} + \frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(y)U(z)\} \\
&\quad + \frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(z)U(x)\} - \mathrm{Tr}\{T^eU^\dagger(x)U(y)T^eU^\dagger(y)U(z)\} \\
&\quad - \mathrm{Tr}\{T^eU^\dagger(x)U(y)T^eU^\dagger(z)U(x)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,x,z)\Big[-\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(x)U(z)\} \\
&\quad + \frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(x)\} - \frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(x)U(z)\}
\end{aligned}
$$

$$
\begin{aligned}
&\quad - \frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(z)U(x)\} + \frac{2}{3}\mathrm{Tr}\{T^eU^\dagger(x)U(y)T^eU^\dagger(z)U(x)\} \\
&\quad - \frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(x)U(z)\} \\
&\quad + \frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w)U(y)\}\,\mathrm{Tr}\{T^eU^\dagger(x)U(z)\} \\
&\quad - \frac{2}{3}\mathrm{Tr}\{T^eU^\dagger(x)U(y)\}\,\mathrm{Tr}\{T^eU^\dagger(x)U(z)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,y,z)\Big[\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(y)U(z)\} \\
&\quad - \frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(y)\} - \frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(y)U(z)\} \\
&\quad - \frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(z)U(y)\} + \frac{2}{3}\mathrm{Tr}\{T^eU^\dagger(x)U(y)T^eU^\dagger(y)U(z)\} \\
&\quad + \frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(y)U(z)\} \\
&\quad - \frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w)U(y)\}\,\mathrm{Tr}\{T^eU^\dagger(y)U(z)\} \\
&\quad + \frac{2}{3}\mathrm{Tr}\{T^eU^\dagger(x)U(y)\}\,\mathrm{Tr}\{T^eU^\dagger(y)U(z)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(x,y,z)\Big[\frac{2N_c}{3}\mathrm{Tr}\{U^\dagger(x)U(z)\} + \frac{2N_c}{3}\mathrm{Tr}\{U^\dagger(z)U(y)\} \\
&\quad - \frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(x)U(z)\} - \frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(y)U(z)\} \\
&\quad - \frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(x)\} - \frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(y)\} \\
&\quad - \frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w)U(x)T^eU^\dagger(x)U(z)\} + \frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(x)T^eU^\dagger(y)U(z)\} \\
&\quad + \frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(x)T^eU^\dagger(z)U(x)\} - \frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w)U(x)T^eU^\dagger(z)U(y)\} \\
&\quad - \frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(x)U(z)\} \\
&\quad + \frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(y)U(z)\} \\
&\quad - \frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(x)U(z)\} \\
&\quad + \frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(y)U(z)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,y,z)\Big[-\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(y)U(z)\} \\
&\quad - \frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w)U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(y)U(z)\}\Big]
\end{aligned}
\tag{4.6}
$$

Group I, Row IV.

$$
\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w',z)\Big[\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(x)U(z)\} \\
&\quad - \mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(z)U(x)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w,z)\Big[-\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(x)U(z)\} \\
&\quad + \mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(x)U(z)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,y,z)\Big[-\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(x)U(z)\} \\
&\quad - \frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(y)\} + \mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(y)U(z)\} \\
&\quad + \mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(z)U(x)\} - \frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(y)U(z)\} \\
&\quad - \frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(z)U(x)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w',z)\Big[\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(y)U(z)\} \\
&\quad - \mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(z)U(y)\}\Big]
\end{aligned}
$$

$$
\begin{aligned}
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w,z)\Big[-\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(y)U(z)\}\\
&\qquad+\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(y)U(z)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,x,z)\Big[-\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(x)U(z)\}\\
&\qquad-\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(x)\}+\frac{1}{3}\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(x)U(z)\}\\
&\qquad+\frac{1}{3}\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(z)U(x)\}-\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(x)U(z)\}\\
&\qquad-\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(z)U(x)\}\\
&\qquad-\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(x)U(z)\}\\
&\qquad-\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w)U(y)\}\,\mathrm{Tr}\{T^eU^\dagger(x)U(z)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,y,z)\Big[-\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(y)U(z)\}\\
&\qquad+\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(y)\}+\frac{1}{3}\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(y)U(z)\}\\
&\qquad+\frac{1}{3}\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(z)U(y)\}-\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(y)U(z)\}\\
&\qquad+\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(z)U(y)\}\\
&\qquad-\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(y)U(z)\}\\
&\qquad-\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(y)\}\,\mathrm{Tr}\{T^eU^\dagger(y)U(z)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(x,y,z)\Big[\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(x)U(z)\}\\
&\qquad+\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(y)U(z)\}+\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(x)\}\\
&\qquad+\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(y)\}-\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(x)U(z)\}\\
&\qquad-\frac{1}{3}\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(y)U(z)\}-\frac{1}{3}\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(z)U(x)\}\\
&\qquad-\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(z)U(y)\}+\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w)U(x)T^eU^\dagger(x)U(z)\}\\
&\qquad+\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(x)T^eU^\dagger(y)U(z)\}+\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(x)T^eU^\dagger(z)U(x)\}\\
&\qquad+\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w)U(x)T^eU^\dagger(z)U(y)\}\\
&\qquad+\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(x)U(z)\}\\
&\qquad-\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(y)U(z)\}\\
&\qquad+\frac{1}{3}\mathrm{Tr}\{T^eU^\dagger(w)U(w')\}\,\mathrm{Tr}\{T^eU^\dagger(x)U(z)\}\\
&\qquad-\frac{1}{3}\mathrm{Tr}\{T^eU^\dagger(w)U(w')\}\,\mathrm{Tr}\{T^eU^\dagger(y)U(z)\}\\
&\qquad-\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(x)U(z)\}\\
&\qquad+\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(y)U(z)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w',z)\Big[-\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(y)U(z)\}\\
&\qquad+\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(z)U(y)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w,z)\Big[\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(y)U(z)\}\\
&\qquad-\mathrm{Tr}\{T^eU^\dagger(w)U(x)T^eU^\dagger(y)U(z)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,y,z)\Big[\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(y)U(z)\}
\end{aligned}
$$

$$+\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w)U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(y)U(z)\}\Big] \tag{4.7}$$

Group I, Row V.

$$\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w',z)\Big[\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(w')\}\\
&\quad-\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(w')U(z)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w,z)\Big[-\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(z)U(w)\}\\
&\quad+\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(z)U(w)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w',z)\Big[\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(w')\}\\
&\quad-\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(w')U(z)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w,z)\Big[-\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(z)U(w)\}\\
&\quad+\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(z)U(w)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(w',w',z)\Big[-2\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(w')\}\\
&\quad+2\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(w')U(z)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(w,w,z)\Big[2\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(z)U(w)\}\\
&\quad-2\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(z)U(w)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w',z)\Big[-\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(w')\}\\
&\quad+\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(w')U(z)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w,z)\Big[\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(z)U(w)\}\\
&\quad-\mathrm{Tr}\{T^eU^\dagger(w)U(x)T^eU^\dagger(z)U(w)\}\Big]
\end{aligned} \tag{4.8}$$

4.2 Group II

Group II, Row I.

$$\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(w,w',z)\Big[2N_c\mathrm{Tr}\{U^\dagger(w)U(w')\}+2N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\quad-2\mathrm{Tr}\{T^eU(w)U^\dagger(z)T^eU(z)U^\dagger(y)\}-2\mathrm{Tr}\{T^eU^\dagger(w')U(z)T^eU^\dagger(z)U(x)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w',z)\Big[-N_c\mathrm{Tr}\{U^\dagger(w)U(w')\}-N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\quad+\mathrm{Tr}\{T^eU(w)U^\dagger(z)T^eU(z)U^\dagger(y)\}+\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(x)U(z)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w,z)\Big[-N_c\mathrm{Tr}\{U^\dagger(w)U(w')\}-N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\quad+\mathrm{Tr}\{T^eU(w')U^\dagger(z)T^eU(z)U^\dagger(x)\}+\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(x)U(z)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,y,z)\Big[N_c\mathrm{Tr}\{U^\dagger(w)U(w')\}+N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\quad-\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(x)U(z)\}-\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(y)\}\\
&\quad-\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(y)U(z)\}-\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(z)U(x)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w',z)\Big[-N_c\mathrm{Tr}\{U^\dagger(w)U(w')\}-N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\quad+\mathrm{Tr}\{T^eU(w)U^\dagger(z)T^eU(z)U^\dagger(y)\}+\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(y)U(z)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w,z)\Big[-N_c\mathrm{Tr}\{U^\dagger(w)U(w')\}-N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\quad+\mathrm{Tr}\{T^eU(w')U^\dagger(z)T^eU(z)U^\dagger(x)\}+\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(y)U(z)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,x,z)\Big[\frac{N_c}{3}\mathrm{Tr}\{U^\dagger(w)U(w')\}+\frac{N_c}{3}\mathrm{Tr}\{U^\dagger(x)U(y)\}
\end{aligned}$$

$$
\begin{aligned}
&- \frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(x)U(z)\} - \frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(x)\}\\
&- \frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(x)U(z)\} - \frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(z)U(x)\}\\
&- \frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(x)U(z)\}\\
&- \frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w)U(y)\}\,\mathrm{Tr}\{T^eU^\dagger(x)U(z)\}\Big]\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,y,z)\Big[\frac{N_c}{3}\mathrm{Tr}\{U^\dagger(w)U(w')\} + \frac{N_c}{3}\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&- \frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(y)U(z)\} + \frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(y)\}\\
&- \frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(y)U(z)\} + \frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(z)U(y)\}\\
&- \frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(y)U(z)\}\\
&- \frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(y)\}\,\mathrm{Tr}\{T^eU^\dagger(y)U(z)\}\Big]\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(x,y,z)\Big[-\frac{4N_c}{3}\mathrm{Tr}\{U^\dagger(w)U(w')\} - \frac{4N_c}{3}\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&+ \frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(x)U(z)\} + \frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(y)U(z)\}\\
&+ \frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(x)\} + \frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(y)\}\\
&+ \frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w)U(x)T^eU^\dagger(x)U(z)\} + \frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(x)T^eU^\dagger(y)U(z)\}\\
&+ \frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(x)T^eU^\dagger(z)U(x)\} + \frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w)U(x)T^eU^\dagger(z)U(y)\}\\
&+ \frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(x)U(z)\}\\
&- \frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(y)U(z)\}\\
&- \frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(x)U(z)\}\\
&+ \frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(y)U(z)\}\Big]\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w',z)\Big[N_c\mathrm{Tr}\{U^\dagger(w)U(w')\} + N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&- \mathrm{Tr}\{T^eU(w)U^\dagger(z)T^eU(z)U^\dagger(x)\} - \mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(y)U(z)\}\Big]\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w,z)\Big[N_c\mathrm{Tr}\{U^\dagger(w)U(w')\} + N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&- \mathrm{Tr}\{T^eU(w')U^\dagger(z)T^eU(z)U^\dagger(x)\} - \mathrm{Tr}\{T^eU^\dagger(w)U(x)T^eU^\dagger(y)U(z)\}\Big]\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,y,z)\Big[\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(y)U(z)\}\\
&+ \frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w)U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(y)U(z)\}\Big] \qquad (4.9)
\end{aligned}
$$

Group II, Row II.

$$
\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,y,z)\Big[N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&- \mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(x)U(y)\} + \mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(y)U(x)\}\\
&- \mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(y)U(x)\}\Big]\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,x,z)\Big[\frac{N_c}{3}\mathrm{Tr}\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\}\Big]\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,y,z)\Big[\frac{N_c}{3}\mathrm{Tr}\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\}\Big]\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(x,y,z)\Big[-\frac{4N_c}{3}\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&+ \frac{1}{3}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(x)U(y)\} + \mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(y)U(x)\}
\end{aligned}
$$

$$
\begin{aligned}
&- \mathrm{Tr}\{T^e U^\dagger(w) U(w') T^e U^\dagger(x) U(y)\} - \frac{1}{3}\mathrm{Tr}\{T^e U^\dagger(w) U(w') T^e U^\dagger(y) U(x)\} \\
&+ \mathrm{Tr}\{T^e U^\dagger(w) U(x) T^e U^\dagger(x) U(y)\} + \frac{1}{3}\mathrm{Tr}\{T^e U^\dagger(w) U(x) T^e U^\dagger(y) U(x)\} \\
&+ \frac{1}{3}\mathrm{Tr}\{T^e U^\dagger(w') U(x)\}\,\mathrm{Tr}\{T^e U^\dagger(x) U(y)\} \\
&+ \frac{1}{3}\mathrm{Tr}\{T^e U^\dagger(w) U(w')\}\,\mathrm{Tr}\{T^e U^\dagger(x) U(y)\} \\
&- \frac{1}{3}\mathrm{Tr}\{T^e U^\dagger(w) U(x)\}\,\mathrm{Tr}\{T^e U^\dagger(x) U(y)\}\Big] \qquad (4.10)
\end{aligned}
$$

Group II, Row III.

$$
\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,y,z)\Big[2N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad - \mathrm{Tr}\{T^e U^\dagger(x) U(y) T^e U^\dagger(y) U(z)\} - \mathrm{Tr}\{T^e U^\dagger(x) U(y) T^e U^\dagger(z) U(x)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,x,z)\Big[- \frac{4N_c}{3}\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad + \frac{2}{3}\mathrm{Tr}\{T^e U^\dagger(x) U(y) T^e U^\dagger(x) U(z)\} + \frac{2}{3}\mathrm{Tr}\{T^e U^\dagger(x) U(y) T^e U^\dagger(z) U(x)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,y,z)\Big[- \frac{4N_c}{3}\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad + \frac{2}{3}\mathrm{Tr}\{T^e U^\dagger(x) U(y) T^e U^\dagger(y) U(z)\} + \frac{2}{3}\mathrm{Tr}\{T^e U^\dagger(x) U(y) T^e U^\dagger(z) U(y)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(x,y,z)\Big[\frac{2N_c\left(N_c^2-1\right)}{3} + \frac{2N_c}{3}\mathrm{Tr}\{U^\dagger(x)U(y)\} - \frac{2N_c}{3}\mathrm{Tr}\{U^\dagger(x)U(z)\} \\
&\quad - \frac{2N_c}{3}\mathrm{Tr}\{U^\dagger(z)U(y)\}\Big] \qquad (4.11)
\end{aligned}
$$

Group II, Row IV.

$$
\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(w,w',z)\Big[2N_c\mathrm{Tr}\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\}\Big] \qquad (4.12)
$$

4.3 Group III (with c.c.)

Group III, Row I.

$$
\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w',z)\Big[N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad - \mathrm{Tr}\{T^e U(w) U^\dagger(x) T^e U(y) U^\dagger(w)\} - \mathrm{Tr}\{T^e U^\dagger(w') U(x) T^e U^\dagger(x) U(z)\} \\
&\quad + \mathrm{Tr}\{T^e U^\dagger(w) U(w') T^e U^\dagger(z) U(x)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w,z)\Big[- N_c\mathrm{Tr}\{U^\dagger(x)U(w')\} + N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad + \mathrm{Tr}\{T^e U^\dagger(w) U(w') T^e U^\dagger(x) U(z)\} - \mathrm{Tr}\{T^e U^\dagger(w) U(y) T^e U^\dagger(x) U(z)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,y,z)\Big[- 2N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad + \mathrm{Tr}\{T^e U^\dagger(w') U(x) T^e U^\dagger(x) U(y)\} + \frac{1}{2}\mathrm{Tr}\{T^e U^\dagger(w') U(x) T^e U^\dagger(x) U(z)\} \\
&\quad + \frac{1}{2}\mathrm{Tr}\{T^e U^\dagger(w') U(x) T^e U^\dagger(z) U(y)\} - \mathrm{Tr}\{T^e U^\dagger(w) U(w') T^e U^\dagger(y) U(z)\} \\
&\quad - \mathrm{Tr}\{T^e U^\dagger(w) U(w') T^e U^\dagger(z) U(x)\} + \mathrm{Tr}\{T^e U^\dagger(w) U(y) T^e U^\dagger(y) U(x)\} \\
&\quad + \frac{1}{2}\mathrm{Tr}\{T^e U^\dagger(w) U(y) T^e U^\dagger(y) U(z)\} + \frac{1}{2}\mathrm{Tr}\{T^e U^\dagger(w) U(y) T^e U^\dagger(z) U(x)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w',z)\Big[- N_c\mathrm{Tr}\{U^\dagger(w)U(y)\} + N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad - \mathrm{Tr}\{T^e U^\dagger(w') U(x) T^e U^\dagger(y) U(z)\} + \mathrm{Tr}\{T^e U^\dagger(w) U(w') T^e U^\dagger(z) U(y)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w,z)\Big[N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad - \mathrm{Tr}\{T^e U(w') U^\dagger(x) T^e U(y) U^\dagger(w')\} + \mathrm{Tr}\{T^e U^\dagger(w) U(w') T^e U^\dagger(y) U(z)\} \\
&\quad - \mathrm{Tr}\{T^e U^\dagger(w) U(y) T^e U^\dagger(y) U(z)\}\Big]
\end{aligned}
$$

$$
\begin{aligned}
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,x,z)\Big[\frac{N_c}{3}\mathrm{Tr}\{U^\dagger(w)U(y)\}+\frac{N_c}{3}\mathrm{Tr}\{U^\dagger(x)U(w')\}\\
&\quad-\frac{2N_c}{3}\mathrm{Tr}\{U^\dagger(x)U(y)\}+\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(x)U(z)\}\\
&\quad+\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(x)\}-\frac{1}{3}\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(x)U(z)\}\\
&\quad-\frac{1}{3}\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(z)U(x)\}+\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(x)U(z)\}\\
&\quad+\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(z)U(x)\}\\
&\quad+\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(x)U(z)\}\\
&\quad+\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w)U(y)\}\,\mathrm{Tr}\{T^eU^\dagger(x)U(z)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,y,z)\Big[\frac{N_c}{3}\mathrm{Tr}\{U^\dagger(w)U(y)\}+\frac{N_c}{3}\mathrm{Tr}\{U^\dagger(x)U(w')\}\\
&\quad-\frac{2N_c}{3}\mathrm{Tr}\{U^\dagger(x)U(y)\}+\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(y)U(z)\}\\
&\quad-\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(y)\}-\frac{1}{3}\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(y)U(z)\}\\
&\quad-\frac{1}{3}\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(z)U(y)\}+\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(y)U(z)\}\\
&\quad-\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(z)U(y)\}\\
&\quad+\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(y)U(z)\}\\
&\quad+\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(y)\}\,\mathrm{Tr}\{T^eU^\dagger(y)U(z)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(x,y,z)\Big[\frac{8N_c}{3}\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\quad-\frac{1}{3}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(x)U(y)\}-\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(x)U(z)\}\\
&\quad-\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(y)U(x)\}-\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(y)U(z)\}\\
&\quad-\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(x)\}-\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(y)\}\\
&\quad+\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(x)U(z)\}+\frac{1}{3}\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(y)U(z)\}\\
&\quad+\frac{1}{3}\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(z)U(x)\}+\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(z)U(y)\}\\
&\quad-\mathrm{Tr}\{T^eU^\dagger(w)U(x)T^eU^\dagger(x)U(y)\}-\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w)U(x)T^eU^\dagger(x)U(z)\}\\
&\quad-\frac{1}{3}\mathrm{Tr}\{T^eU^\dagger(w)U(x)T^eU^\dagger(y)U(x)\}-\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(x)T^eU^\dagger(y)U(z)\}\\
&\quad-\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(x)T^eU^\dagger(z)U(x)\}-\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w)U(x)T^eU^\dagger(z)U(y)\}\\
&\quad-\frac{1}{3}\mathrm{Tr}\{T^eU^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(x)U(y)\}\\
&\quad-\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(x)U(z)\}\\
&\quad+\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(y)U(z)\}\\
&\quad-\frac{1}{3}\mathrm{Tr}\{T^eU^\dagger(w)U(w')\}\,\mathrm{Tr}\{T^eU^\dagger(x)U(z)\}\\
&\quad+\frac{1}{3}\mathrm{Tr}\{T^eU^\dagger(w)U(w')\}\,\mathrm{Tr}\{T^eU^\dagger(y)U(z)\}\\
&\quad+\frac{1}{3}\mathrm{Tr}\{T^eU^\dagger(w)U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(x)U(y)\}\\
&\quad+\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(x)U(z)\}\\
&\quad-\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(y)U(z)\}\Big]
\end{aligned}
$$

$$
\begin{aligned}
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w',z)\Big[-N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad+\mathrm{Tr}\{T^eU(w)U^\dagger(x)T^eU(y)U^\dagger(w)\}+\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(y)U(z)\}\\
&\qquad-\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(z)U(y)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w,z)\Big[-N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad+\mathrm{Tr}\{T^eU(w')U^\dagger(x)T^eU(y)U^\dagger(w')\}-\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(y)U(z)\}\\
&\qquad+\mathrm{Tr}\{T^eU^\dagger(w)U(x)T^eU^\dagger(y)U(z)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,y,z)\Big[-\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(y)U(z)\}\\
&\qquad-\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w)U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(y)U(z)\}\Big]
\end{aligned}
\tag{4.13}
$$

Group III, Row II.

$$
\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w',z)\Big[-N_c\mathrm{Tr}\{U^\dagger(x)U(w')\}-N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad+\mathrm{Tr}\{T^eU(x)U^\dagger(z)T^eU(z)U^\dagger(y)\}+\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(x)U(z)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w,z)\Big[N_c\mathrm{Tr}\{U^\dagger(w)U(y)\}+N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad-\mathrm{Tr}\{T^eU(x)U^\dagger(z)T^eU(z)U^\dagger(y)\}-\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(x)U(z)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,y,z)\Big[-\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(x)U(z)\}\\
&\qquad+\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(y)\}-\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(y)U(z)\}\\
&\qquad+\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(z)U(x)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w',z)\Big[N_c\mathrm{Tr}\{U^\dagger(x)U(w')\}+N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad-\mathrm{Tr}\{T^eU(x)U^\dagger(z)T^eU(z)U^\dagger(y)\}-\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(y)U(z)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w,z)\Big[-N_c\mathrm{Tr}\{U^\dagger(w)U(y)\}-N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad+\mathrm{Tr}\{T^eU(x)U^\dagger(z)T^eU(z)U^\dagger(y)\}+\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(y)U(z)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,x,z)\Big[-\frac{N_c}{3}\mathrm{Tr}\{U^\dagger(w)U(y)\}+\frac{2N_c}{3}\mathrm{Tr}\{U^\dagger(x)U(w')\}\\
&\qquad+\frac{N_c}{3}\mathrm{Tr}\{U^\dagger(x)U(y)\}-\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(x)U(z)\}\\
&\qquad-\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(x)\}+\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(x)U(z)\}\\
&\qquad+\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(z)U(x)\}-\frac{1}{3}\mathrm{Tr}\{T^eU^\dagger(x)U(y)T^eU^\dagger(z)U(x)\}\\
&\qquad-\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(x)U(z)\}\\
&\qquad+\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w)U(y)\}\,\mathrm{Tr}\{T^eU^\dagger(x)U(z)\}\\
&\qquad-\frac{2}{3}\mathrm{Tr}\{T^eU^\dagger(x)U(y)\}\,\mathrm{Tr}\{T^eU^\dagger(x)U(z)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,y,z)\Big[\frac{2N_c}{3}\mathrm{Tr}\{U^\dagger(w)U(y)\}-\frac{N_c}{3}\mathrm{Tr}\{U^\dagger(x)U(w')\}\\
&\qquad+\frac{N_c}{3}\mathrm{Tr}\{U^\dagger(x)U(y)\}+\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(y)U(z)\}\\
&\qquad-\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)T^eU^\dagger(z)U(y)\}-\frac{1}{2}\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(y)U(z)\}\\
&\qquad-\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(z)U(y)\}+\frac{1}{3}\mathrm{Tr}\{T^eU^\dagger(x)U(y)T^eU^\dagger(y)U(z)\}\\
&\qquad-\frac{2}{3}\mathrm{Tr}\{T^eU^\dagger(x)U(y)T^eU^\dagger(z)U(y)\}\\
&\qquad+\frac{1}{6}\mathrm{Tr}\{T^eU^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(y)U(z)\}
\end{aligned}
$$

$$
\begin{aligned}
&\quad + \frac{1}{6}\mathrm{Tr}\{T^e U^\dagger(w)U(y)\}\,\mathrm{Tr}\{T^e U^\dagger(y)U(z)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(x,y,z)\Big[\frac{N_c}{3}\mathrm{Tr}\{U^\dagger(x)U(w')\} + \frac{N_c}{3}\mathrm{Tr}\{U^\dagger(x)U(w)\} \\
&\quad + \frac{2N_c}{3}\mathrm{Tr}\{U^\dagger(x)U(y)\} - \frac{4N_c}{3}\mathrm{Tr}\{U^\dagger(x)U(z)\} + \frac{2N_c}{3}\mathrm{Tr}\{U^\dagger(z)U(y)\} \\
&\quad + \frac{1}{6}\mathrm{Tr}\{T^e U^\dagger(w')U(x)T^e U^\dagger(x)U(z)\} - \frac{1}{2}\mathrm{Tr}\{T^e U^\dagger(w')U(x)T^e U^\dagger(y)U(z)\} \\
&\quad + \frac{1}{6}\mathrm{Tr}\{T^e U^\dagger(w')U(x)T^e U^\dagger(z)U(x)\} - \frac{1}{6}\mathrm{Tr}\{T^e U^\dagger(w')U(x)T^e U^\dagger(z)U(y)\} \\
&\quad + \frac{1}{2}\mathrm{Tr}\{T^e U^\dagger(w)U(x)T^e U^\dagger(x)U(z)\} - \frac{1}{6}\mathrm{Tr}\{T^e U^\dagger(w)U(x)T^e U^\dagger(y)U(z)\} \\
&\quad - \frac{1}{6}\mathrm{Tr}\{T^e U^\dagger(w)U(x)T^e U^\dagger(z)U(x)\} - \frac{1}{2}\mathrm{Tr}\{T^e U^\dagger(w)U(x)T^e U^\dagger(z)U(y)\} \\
&\quad + \frac{1}{2}\mathrm{Tr}\{T^e U^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^e U^\dagger(x)U(z)\} \\
&\quad + \frac{1}{6}\mathrm{Tr}\{T^e U^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^e U^\dagger(y)U(z)\} \\
&\quad + \frac{1}{6}\mathrm{Tr}\{T^e U^\dagger(w)U(x)\}\,\mathrm{Tr}\{T^e U^\dagger(x)U(z)\} \\
&\quad - \frac{1}{6}\mathrm{Tr}\{T^e U^\dagger(w)U(x)\}\,\mathrm{Tr}\{T^e U^\dagger(y)U(z)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w',z)\Big[- N_c\mathrm{Tr}\{U^\dagger(x)U(w')\} - N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad + N_c\mathrm{Tr}\{U^\dagger(x)U(z)\} + \mathrm{Tr}\{T^e U^\dagger(w')U(x)T^e U^\dagger(y)U(z)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w,z)\Big[- N_c\mathrm{Tr}\{U^\dagger(x)U(w)\} - N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad + N_c\mathrm{Tr}\{U^\dagger(x)U(z)\} + \mathrm{Tr}\{T^e U^\dagger(w)U(x)T^e U^\dagger(y)U(z)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,y,z)\Big[- \frac{1}{2}\mathrm{Tr}\{T^e U^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^e U^\dagger(y)U(z)\} \\
&\quad - \frac{1}{2}\mathrm{Tr}\{T^e U^\dagger(w)U(x)\}\,\mathrm{Tr}\{T^e U^\dagger(y)U(z)\}\Big] \qquad (4.14)
\end{aligned}
$$

Group III, Row III.

$$
\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(w,w',z)\Big[2N_c\mathrm{Tr}\{U^\dagger(w)U(y)\} + 2N_c\mathrm{Tr}\{U^\dagger(x)U(w')\} \\
&\quad - 4N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} - 2\mathrm{Tr}\{T^e U(w)U^\dagger(z)T^e U(z)U^\dagger(w')\} \\
&\quad + 2\mathrm{Tr}\{T^e U(w)U^\dagger(z)T^e U(z)U^\dagger(y)\} + 2\mathrm{Tr}\{T^e U^\dagger(w')U(z)T^e U^\dagger(z)U(x)\} \\
&\quad - 2\mathrm{Tr}\{T^e U^\dagger(w)U(z)T^e U^\dagger(z)U(w')\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w',z)\Big[- N_c\mathrm{Tr}\{U^\dagger(x)U(w')\} + N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad + \mathrm{Tr}\{T^e U(w)U^\dagger(z)T^e U(z)U^\dagger(w')\} - \mathrm{Tr}\{T^e U(w)U^\dagger(z)T^e U(z)U^\dagger(y)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w,z)\Big[N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad - \mathrm{Tr}\{T^e U(w')U^\dagger(z)T^e U(z)U^\dagger(x)\} - \mathrm{Tr}\{T^e U(w)U^\dagger(y)T^e U(y)U^\dagger(x)\} \\
&\quad + \mathrm{Tr}\{T^e U(w)U^\dagger(z)T^e U(z)U^\dagger(w')\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w',z)\Big[N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad - \mathrm{Tr}\{T^e U(w')U^\dagger(x)T^e U(x)U^\dagger(y)\} + \mathrm{Tr}\{T^e U(w)U^\dagger(z)T^e U(z)U^\dagger(w')\} \\
&\quad - \mathrm{Tr}\{T^e U(w)U^\dagger(z)T^e U(z)U^\dagger(y)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w,z)\Big[- N_c\mathrm{Tr}\{U^\dagger(w)U(y)\} + N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad - \mathrm{Tr}\{T^e U(w')U^\dagger(x)T^e U(x)U^\dagger(z)\} + \mathrm{Tr}\{T^e U(w)U^\dagger(z)T^e U(z)U^\dagger(w')\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w',z)\Big[- N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad + \mathrm{Tr}\{T^e U(w')U^\dagger(x)T^e U(x)U^\dagger(y)\} - \mathrm{Tr}\{T^e U(w)U^\dagger(z)T^e U(z)U^\dagger(w')\}
\end{aligned}
$$

$$
\begin{aligned}
&\quad + \mathrm{Tr}\{T^e U(w)U^\dagger(z)T^e U(z)U^\dagger(x)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w,z)\Big[- N_c \mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad + \mathrm{Tr}\{T^e U(w')U^\dagger(z)T^e U(z)U^\dagger(x)\} + \mathrm{Tr}\{T^e U(w)U^\dagger(x)T^e U(x)U^\dagger(y)\} \\
&\quad - \mathrm{Tr}\{T^e U(w)U^\dagger(z)T^e U(z)U^\dagger(w')\}\Big]
\end{aligned} \tag{4.15}
$$

Group III, Row IV.

$$
\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,y,z)\Big[\frac{1}{2}\mathrm{Tr}\{T^e U^\dagger(w')U(x)T^e U^\dagger(x)U(z)\} \\
&\quad - \frac{1}{2}\mathrm{Tr}\{T^e U^\dagger(w')U(x)T^e U^\dagger(z)U(y)\} + \frac{1}{2}\mathrm{Tr}\{T^e U^\dagger(w)U(y)T^e U^\dagger(y)U(z)\} \\
&\quad - \frac{1}{2}\mathrm{Tr}\{T^e U^\dagger(w)U(y)T^e U^\dagger(z)U(x)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,x,z)\Big[- \frac{N_c}{3}\mathrm{Tr}\{U^\dagger(w)U(y)\} + \frac{2N_c}{3}\mathrm{Tr}\{U^\dagger(x)U(w')\} \\
&\quad - \frac{N_c}{3}\mathrm{Tr}\{U^\dagger(x)U(y)\} - \frac{1}{2}\mathrm{Tr}\{T^e U^\dagger(w')U(x)T^e U^\dagger(x)U(z)\} \\
&\quad - \frac{1}{6}\mathrm{Tr}\{T^e U^\dagger(w')U(x)T^e U^\dagger(z)U(x)\} + \frac{1}{6}\mathrm{Tr}\{T^e U^\dagger(w)U(y)T^e U^\dagger(x)U(z)\} \\
&\quad + \frac{1}{6}\mathrm{Tr}\{T^e U^\dagger(w)U(y)T^e U^\dagger(z)U(x)\} + \frac{1}{3}\mathrm{Tr}\{T^e U^\dagger(x)U(y)T^e U^\dagger(z)U(x)\} \\
&\quad + \frac{1}{6}\mathrm{Tr}\{T^e U^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^e U^\dagger(x)U(z)\} \\
&\quad - \frac{1}{2}\mathrm{Tr}\{T^e U^\dagger(w)U(y)\}\,\mathrm{Tr}\{T^e U^\dagger(x)U(z)\} \\
&\quad + \frac{2}{3}\mathrm{Tr}\{T^e U^\dagger(x)U(y)\}\,\mathrm{Tr}\{T^e U^\dagger(x)U(z)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,y,z)\Big[\frac{2N_c}{3}\mathrm{Tr}\{U^\dagger(w)U(y)\} - \frac{N_c}{3}\mathrm{Tr}\{U^\dagger(x)U(w')\} \\
&\quad - \frac{N_c}{3}\mathrm{Tr}\{U^\dagger(x)U(y)\} - \frac{1}{6}\mathrm{Tr}\{T^e U^\dagger(w')U(x)T^e U^\dagger(y)U(z)\} \\
&\quad + \frac{1}{2}\mathrm{Tr}\{T^e U^\dagger(w')U(x)T^e U^\dagger(z)U(y)\} - \frac{1}{6}\mathrm{Tr}\{T^e U^\dagger(w)U(y)T^e U^\dagger(y)U(z)\} \\
&\quad - \frac{1}{2}\mathrm{Tr}\{T^e U^\dagger(w)U(y)T^e U^\dagger(z)U(y)\} - \frac{1}{3}\mathrm{Tr}\{T^e U^\dagger(x)U(y)T^e U^\dagger(y)U(z)\} \\
&\quad + \frac{2}{3}\mathrm{Tr}\{T^e U^\dagger(x)U(y)T^e U^\dagger(z)U(y)\} \\
&\quad - \frac{1}{6}\mathrm{Tr}\{T^e U^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^e U^\dagger(y)U(z)\} \\
&\quad - \frac{1}{6}\mathrm{Tr}\{T^e U^\dagger(w)U(y)\}\,\mathrm{Tr}\{T^e U^\dagger(y)U(z)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(x,y,z)\Big[- \frac{2N_c}{3}\mathrm{Tr}\{U^\dagger(x)U(y)\} + \frac{4N_c}{3}\mathrm{Tr}\{U^\dagger(x)U(z)\} \\
&\quad - \frac{2N_c}{3}\mathrm{Tr}\{U^\dagger(z)U(y)\} - \frac{1}{6}\mathrm{Tr}\{T^e U^\dagger(w')U(x)T^e U^\dagger(x)U(z)\} \\
&\quad + \frac{1}{3}\mathrm{Tr}\{T^e U^\dagger(w')U(x)T^e U^\dagger(y)U(x)\} + \frac{1}{6}\mathrm{Tr}\{T^e U^\dagger(w')U(x)T^e U^\dagger(y)U(z)\} \\
&\quad - \frac{1}{2}\mathrm{Tr}\{T^e U^\dagger(w')U(x)T^e U^\dagger(z)U(x)\} + \frac{1}{6}\mathrm{Tr}\{T^e U^\dagger(w')U(x)T^e U^\dagger(z)U(y)\} \\
&\quad - \frac{1}{2}\mathrm{Tr}\{T^e U^\dagger(w)U(x)T^e U^\dagger(x)U(z)\} + \frac{1}{3}\mathrm{Tr}\{T^e U^\dagger(w)U(x)T^e U^\dagger(y)U(x)\} \\
&\quad - \frac{1}{6}\mathrm{Tr}\{T^e U^\dagger(w)U(x)T^e U^\dagger(y)U(z)\} - \frac{1}{6}\mathrm{Tr}\{T^e U^\dagger(w)U(x)T^e U^\dagger(z)U(x)\} \\
&\quad + \frac{1}{2}\mathrm{Tr}\{T^e U^\dagger(w)U(x)T^e U^\dagger(z)U(y)\} \\
&\quad - \frac{1}{3}\mathrm{Tr}\{T^e U^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^e U^\dagger(x)U(y)\} \\
&\quad - \frac{1}{6}\mathrm{Tr}\{T^e U^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^e U^\dagger(x)U(z)\} \\
&\quad - \frac{1}{2}\mathrm{Tr}\{T^e U^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^e U^\dagger(y)U(z)\}
\end{aligned}
$$

$$
\begin{aligned}
&\quad - \frac{1}{3}\mathrm{Tr}\{T^e U^\dagger(w)U(x)\}\,\mathrm{Tr}\{T^e U^\dagger(x)U(y)\} \\
&\quad + \frac{1}{6}\mathrm{Tr}\{T^e U^\dagger(w)U(x)\}\,\mathrm{Tr}\{T^e U^\dagger(x)U(z)\} \\
&\quad - \frac{1}{6}\mathrm{Tr}\{T^e U^\dagger(w)U(x)\}\,\mathrm{Tr}\{T^e U^\dagger(y)U(z)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,y,z)\Big[\frac{1}{2}\mathrm{Tr}\{T^e U^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^e U^\dagger(y)U(z)\} \\
&\quad + \frac{1}{2}\mathrm{Tr}\{T^e U^\dagger(w)U(x)\}\,\mathrm{Tr}\{T^e U^\dagger(y)U(z)\}\Big] \qquad (4.16)
\end{aligned}
$$

Group III, Row V.

$$
\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w',z)\Big[N_c\mathrm{Tr}\{U^\dagger(x)U(w')\} - N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad - \mathrm{Tr}\{T^e U(w)U^\dagger(w')T^e U(x)U^\dagger(w)\} + \mathrm{Tr}\{T^e U(w)U^\dagger(x)T^e U(y)U^\dagger(w)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w,z)\Big[N_c\mathrm{Tr}\{U^\dagger(x)U(w')\} - N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad - \mathrm{Tr}\{T^e U(w)U^\dagger(w')T^e U(w')U^\dagger(x)\} + \mathrm{Tr}\{T^e U(w)U^\dagger(y)T^e U(y)U^\dagger(x)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w',z)\Big[N_c\mathrm{Tr}\{U^\dagger(w)U(y)\} - N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad + \mathrm{Tr}\{T^e U(w')U^\dagger(x)T^e U(x)U^\dagger(y)\} - \mathrm{Tr}\{T^e U(w)U^\dagger(w')T^e U(y)U^\dagger(w)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w,z)\Big[N_c\mathrm{Tr}\{U^\dagger(w)U(y)\} - N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad + \mathrm{Tr}\{T^e U(w')U^\dagger(x)T^e U(y)U^\dagger(w')\} - \mathrm{Tr}\{T^e U(w)U^\dagger(w')T^e U(w')U^\dagger(y)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w',z)\Big[N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad - \mathrm{Tr}\{T^e U(w')U^\dagger(x)T^e U(x)U^\dagger(y)\} + \mathrm{Tr}\{T^e U(w)U^\dagger(w')T^e U(y)U^\dagger(w)\} \\
&\quad - \mathrm{Tr}\{T^e U(w)U^\dagger(x)T^e U(y)U^\dagger(w)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w,z)\Big[N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad - \mathrm{Tr}\{T^e U(w')U^\dagger(x)T^e U(y)U^\dagger(w')\} + \mathrm{Tr}\{T^e U(w)U^\dagger(w')T^e U(w')U^\dagger(y)\} \\
&\quad - \mathrm{Tr}\{T^e U(w)U^\dagger(x)T^e U(x)U^\dagger(y)\}\Big] \qquad (4.17)
\end{aligned}
$$

Group III, Row VI.

$$
\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w',z)\Big[- N_c\mathrm{Tr}\{U^\dagger(x)U(w')\} + N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad + \mathrm{Tr}\{T^e U(w')U^\dagger(z)T^e U(z)U^\dagger(x)\} - \mathrm{Tr}\{T^e U(x)U^\dagger(z)T^e U(z)U^\dagger(y)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w,z)\Big[- N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad + \mathrm{Tr}\{T^e U(w)U^\dagger(y)T^e U(y)U^\dagger(x)\} - \mathrm{Tr}\{T^e U(w)U^\dagger(z)T^e U(z)U^\dagger(y)\} \\
&\quad + \mathrm{Tr}\{T^e U(x)U^\dagger(z)T^e U(z)U^\dagger(y)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w',z)\Big[- N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad + \mathrm{Tr}\{T^e U(w')U^\dagger(x)T^e U(x)U^\dagger(y)\} - \mathrm{Tr}\{T^e U(w')U^\dagger(z)T^e U(z)U^\dagger(x)\} \\
&\quad + \mathrm{Tr}\{T^e U(x)U^\dagger(z)T^e U(z)U^\dagger(y)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w,z)\Big[- N_c\mathrm{Tr}\{U^\dagger(w)U(y)\} + N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad + \mathrm{Tr}\{T^e U(w)U^\dagger(z)T^e U(z)U^\dagger(y)\} - \mathrm{Tr}\{T^e U(x)U^\dagger(z)T^e U(z)U^\dagger(y)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w',z)\Big[N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} - N_c\mathrm{Tr}\{U^\dagger(x)U(z)\} \\
&\quad - \mathrm{Tr}\{T^e U(w')U^\dagger(x)T^e U(x)U^\dagger(y)\} + \mathrm{Tr}\{T^e U(w')U^\dagger(z)T^e U(z)U^\dagger(x)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w,z)\Big[N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} - N_c\mathrm{Tr}\{U^\dagger(x)U(z)\} \\
&\quad - \mathrm{Tr}\{T^e U(w)U^\dagger(x)T^e U(x)U^\dagger(y)\} + \mathrm{Tr}\{T^e U(w)U^\dagger(z)T^e U(z)U^\dagger(x)\}\Big] \qquad (4.18)
\end{aligned}
$$

A Your lists in singlet form, exactly as written

Here every term of your lists is projected and written as traces with the index order exactly as in your files. Where your index order runs against the chain, the Wilson line appears as $U^\dagger$ (this is exact: $U^T = U^\dagger$).

- With your convention $U^\dagger = U$, each bracket can be simplified further, and the sum of all of them is Eq. (3.1) (Table 1).
- For a general U it is not.
- Same normalization, Eq. (2.1). "Genuine" is Sec. 1 of `Evolution_terms.tex` (the dimensionally regulated brackets written back as the z integrals they come from).

A.1 Genuine

Group I, Row II.

$$\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w,z)\Big[N_c\mathrm{Tr}\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w',z)\Big[N_c\mathrm{Tr}\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(w',w',z)\Big[-N_c\mathrm{Tr}\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(w,w,z)\Big[-N_c\mathrm{Tr}\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\}\Big]
\end{aligned}\tag{A.1}$$

Group I, Row V.

$$\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w,z)\Big[N_c\mathrm{Tr}\{U^\dagger(x)U(w')\}-N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad-\mathrm{Tr}\{T^eU(w)U^\dagger(w')T^eU(w')U^\dagger(z)\}+\mathrm{Tr}\{T^eU(w)U^\dagger(y)T^eU(y)U^\dagger(z)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w',z)\Big[N_c\mathrm{Tr}\{U^\dagger(w)U(y)\}-N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad+\mathrm{Tr}\{T^eU(w')U^\dagger(x)T^eU(x)U^\dagger(z)\}-\mathrm{Tr}\{T^eU(w)U^\dagger(w')T^eU(z)U^\dagger(w)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(w',w',z)\Big[-2N_c\mathrm{Tr}\{U^\dagger(w)U(y)\}+2N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad-2\mathrm{Tr}\{T^eU(w')U^\dagger(x)T^eU(x)U^\dagger(z)\}+2\mathrm{Tr}\{T^eU(w)U^\dagger(w')T^eU(z)U^\dagger(w)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(w,w,z)\Big[-2N_c\mathrm{Tr}\{U^\dagger(x)U(w')\}+2N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad+2\mathrm{Tr}\{T^eU(w)U^\dagger(w')T^eU(w')U^\dagger(z)\}-2\mathrm{Tr}\{T^eU(w)U^\dagger(y)T^eU(y)U^\dagger(z)\}\Big]
\end{aligned}\tag{A.2}$$

Group II, Row I.

$$\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(w,w',z)\Big[2N_c\mathrm{Tr}\{U^\dagger(w)U(w')\}+2N_c\mathrm{Tr}\{U(x)U(y)\}\\
&\qquad-2\mathrm{Tr}\{T^eU(w)U(y)T^eU^\dagger(z)U^\dagger(w)\}-2\mathrm{Tr}\{T^eU(z)U(w')T^eU^\dagger(w')U(x)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w',z)\Big[-N_c\mathrm{Tr}\{U^\dagger(w)U(w')\}-N_c\mathrm{Tr}\{U(x)U(y)\}\\
&\qquad+\mathrm{Tr}\{T^eU(w)U(y)T^eU^\dagger(z)U^\dagger(w)\}+\mathrm{Tr}\{T^eU(z)U(w')T^eU^\dagger(w')U(x)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,y,z)\Big[\frac{N_c}{2}\mathrm{Tr}\{U^\dagger(w)U(w')\}+\frac{N_c}{2}\mathrm{Tr}\{U(x)U(y)\}\\
&\qquad-\frac{1}{2}\mathrm{Tr}\{T^eU(w)U(y)T^eU^\dagger(z)U^\dagger(w)\}-\frac{1}{2}\mathrm{Tr}\{T^eU(z)U(w')T^eU^\dagger(w')U(x)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w,z)\Big[-N_c\mathrm{Tr}\{U^\dagger(w)U(w')\}-N_c\mathrm{Tr}\{U(x)U(y)\}\\
&\qquad+\mathrm{Tr}\{T^eU(w)U(y)T^eU^\dagger(z)U^\dagger(w)\}+\mathrm{Tr}\{T^eU(z)U(w')T^eU^\dagger(w')U(x)\}\Big]
\end{aligned}\tag{A.3}$$

Group II, Row IV.

$$\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(w,w',z)\Big[2N_c\mathrm{Tr}\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(w',w',z)\Big[-N_c\mathrm{Tr}\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(w,w,z)\Big[-N_c\mathrm{Tr}\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\}\Big]
\end{aligned}\tag{A.4}$$

Group III, Row III.

$$
\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(w,w',z)\Big[-4N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\quad-2\mathrm{Tr}\{T^eU(w')U(w)T^eU^\dagger(w)U^\dagger(z)\}+2\mathrm{Tr}\{T^eU(w')U^\dagger(x)T^eU(x)U(z)\}\\
&\quad-2\mathrm{Tr}\{T^eU(w)U(w')T^eU^\dagger(w')U^\dagger(z)\}+2\mathrm{Tr}\{T^eU(w)U^\dagger(y)T^eU(y)U(z)\}\\
&\quad+2\mathrm{Tr}\{T^eU(x)U(w')T^eU(z)U(z)\}+2\mathrm{Tr}\{T^eU(y)U(w)T^eU(z)U(z)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w',z)\Big[N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\quad+\mathrm{Tr}\{T^eU(w)U(w')T^eU^\dagger(w')U^\dagger(z)\}-\mathrm{Tr}\{T^eU(w)U^\dagger(y)T^eU(y)U(z)\}\\
&\quad-\mathrm{Tr}\{T^eU(x)U(w')T^eU(z)U(z)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w,z)\Big[N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\quad+\mathrm{Tr}\{T^eU(w')U(w)T^eU^\dagger(w)U^\dagger(z)\}-\mathrm{Tr}\{T^eU(w')U^\dagger(x)T^eU(x)U(z)\}\\
&\quad-\mathrm{Tr}\{T^eU(y)U(w)T^eU(z)U(z)\}\Big]
\end{aligned}
\tag{A.5}
$$

A.2 From four ρ's

`4rho.tex` **lists (missing in** `Evolution_terms.tex`**, M1).**

$$
\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,y,z)\Big[\frac{N_c}{2}\mathrm{Tr}\{U(w)U(w')\}-N_c\mathrm{Tr}\{U^\dagger(w)U(y)\}\\
&\quad-N_c\mathrm{Tr}\{U^\dagger(x)U(w')\}+\frac{7N_c}{2}\mathrm{Tr}\{U^\dagger(x)U(y)\}+\frac{1}{2}\mathrm{Tr}\{T^eU(w')U^\dagger(z)T^eU(z)U^\dagger(x)\}\\
&\quad-2\mathrm{Tr}\{T^eU(x)U(z)T^eU^\dagger(z)U(y)\}-\mathrm{Tr}\{T^eU(y)U(x)T^eU^\dagger(x)U(w)\}\\
&\quad+\frac{1}{2}\mathrm{Tr}\{T^eU(y)U^\dagger(z)T^eU(z)U(w)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,x,z)\Big[2N_c\mathrm{Tr}\{U^\dagger(x)U(w')\}-2N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\quad+2\mathrm{Tr}\{T^eU(w')U^\dagger(x)T^eU(z)U^\dagger(x)\}-3\mathrm{Tr}\{T^eU(w)U^\dagger(w')T^eU(x)U^\dagger(z)\}\\
&\quad+3\mathrm{Tr}\{T^eU(w)U^\dagger(w')T^eU(z)U^\dagger(x)\}-3\mathrm{Tr}\{T^eU(x)U(w')T^eU^\dagger(z)U^\dagger(x)\}\\
&\quad+2\mathrm{Tr}\{T^eU(x)U^\dagger(y)T^eU(x)U^\dagger(z)\}-\mathrm{Tr}\{T^eU(x)U^\dagger(z)T^eU(z)U(y)\}\\
&\quad-\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(x)U(z)\}+\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(z)U(x)\}\\
&\quad-3\mathrm{Tr}\{T^eU(w)U(w')\}\,\mathrm{Tr}\{T^eU(z)U(x)\}\\
&\quad+2\mathrm{Tr}\{T^eU^\dagger(w')U(x)\}\,\mathrm{Tr}\{T^eU^\dagger(x)U(z)\}\\
&\quad+\mathrm{Tr}\{T^eU^\dagger(w)U(y)\}\,\mathrm{Tr}\{T^eU^\dagger(x)U(z)\}\\
&\quad-2\mathrm{Tr}\{T^eU^\dagger(x)U(y)\}\,\mathrm{Tr}\{T^eU^\dagger(x)U(z)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,y,z)\Big[2N_c\mathrm{Tr}\{U^\dagger(w)U(y)\}-2N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\quad-\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^eU^\dagger(z)U(y)\}+\mathrm{Tr}\{T^eU^\dagger(x)U(y)T^eU^\dagger(z)U(y)\}\\
&\quad-\mathrm{Tr}\{T^eU^\dagger(w)U(y)\}\,\mathrm{Tr}\{T^eU^\dagger(y)U(z)\}\\
&\quad+\mathrm{Tr}\{T^eU^\dagger(x)U(y)\}\,\mathrm{Tr}\{T^eU^\dagger(y)U(z)\}\Big]
\end{aligned}
\tag{A.6}
$$

your Sec. 2.

$$
\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,y,z)\Big[\frac{N_c}{2}\mathrm{Tr}\{U(w')U(x)\}-N_c\mathrm{Tr}\{U(w)U(w')\}\\
&\quad+\frac{N_c}{2}\mathrm{Tr}\{U(w)U(y)\}+\frac{1}{2}\mathrm{Tr}\{T^eU(x)U(w)T^eU^\dagger(y)U^\dagger(x)\}\\
&\quad-\frac{1}{2}\mathrm{Tr}\{T^eU(y)U(w')T^eU^\dagger(x)U^\dagger(y)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,x,z)\Big[-\frac{N_c}{2}\mathrm{Tr}\{U(w')U(x)\}-\frac{N_c}{2}\mathrm{Tr}\{U(w)U(w')\}\\
&\quad+\frac{N_c}{2}\mathrm{Tr}\{U(w)U(y)\}+\frac{N_c}{2}\mathrm{Tr}\{U^\dagger(x)U(y)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,y,z)\Big[\frac{N_c}{2}\mathrm{Tr}\{U(w')U(x)\}-\frac{N_c}{2}\mathrm{Tr}\{U(w)U(y)\}
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{N_c}{2}\mathrm{Tr}\{U^\dagger(w)U(w')\} + \frac{N_c}{2}\mathrm{Tr}\{U^\dagger(x)U(y)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(x,y,z)\Big[- \frac{N_c}{2}\mathrm{Tr}\{U(w')U(x)\} + \frac{N_c}{2}\mathrm{Tr}\{U(w)U(w')\} \\
&- \frac{N_c}{2}\mathrm{Tr}\{U(w)U(x)\} + \frac{N_c}{2}\mathrm{Tr}\{U^\dagger(w)U(w')\} - N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&+ \frac{1}{2}\mathrm{Tr}\{T^eU(y)U(w')T^eU^\dagger(x)U^\dagger(y)\} + \frac{1}{2}\mathrm{Tr}\{T^eU(y)U(w)T^eU^\dagger(x)U^\dagger(y)\}\Big]
\end{aligned} \tag{A.7}
$$

A.3 From three ρ's

Group I, Row II.

$$
\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w',z)\Big[- N_c\mathrm{Tr}\{U(w')U(x)\} + N_c\mathrm{Tr}\{U(w)U(w')\} \\
&\quad + N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} - \mathrm{Tr}\{T^eU(w)U^\dagger(x)T^eU(y)U^\dagger(w)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,y,z)\Big[- 2N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad + \mathrm{Tr}\{T^eU(w')U^\dagger(x)T^eU(y)U^\dagger(w')\} + \mathrm{Tr}\{T^eU(w)U^\dagger(x)T^eU(y)U^\dagger(w)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w,z)\Big[N_c\mathrm{Tr}\{U(w)U(w')\} - N_c\mathrm{Tr}\{U(w)U(y)\} \\
&\quad + N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} - \mathrm{Tr}\{T^eU(w')U^\dagger(x)T^eU(y)U^\dagger(w')\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(x,y,z)\Big[2N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad - \mathrm{Tr}\{T^eU(w')U^\dagger(x)T^eU(y)U^\dagger(w')\} - \mathrm{Tr}\{T^eU(w)U^\dagger(x)T^eU(y)U^\dagger(w)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w',z)\Big[N_c\mathrm{Tr}\{U(w')U(x)\} - N_c\mathrm{Tr}\{U(w)U(w')\} \\
&\quad - N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} + \mathrm{Tr}\{T^eU(w)U^\dagger(x)T^eU(y)U^\dagger(w)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w,z)\Big[- N_c\mathrm{Tr}\{U(w)U(w')\} + N_c\mathrm{Tr}\{U(w)U(x)\} \\
&\quad - N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} + \mathrm{Tr}\{T^eU(w')U^\dagger(x)T^eU(y)U^\dagger(w')\}\Big]
\end{aligned} \tag{A.8}
$$

Group I, Row III, N_1.

$$
\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,y,z)\Big[N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad - \frac{1}{2}\mathrm{Tr}\{T^eU(w')U^\dagger(x)T^eU(y)U^\dagger(w')\} + \frac{1}{2}\mathrm{Tr}\{T^eU(w')U^\dagger(x)T^eU(z)U^\dagger(w')\} \\
&\quad - \frac{1}{2}\mathrm{Tr}\{T^eU(w)U^\dagger(x)T^eU(y)U^\dagger(w)\} + \frac{1}{2}\mathrm{Tr}\{T^eU(w)U^\dagger(y)T^eU(z)U^\dagger(w)\} \\
&\quad - \frac{1}{2}\mathrm{Tr}\{T^eU(x)U^\dagger(y)T^eU(y)U^\dagger(z)\} - \frac{1}{2}\mathrm{Tr}\{T^eU(x)U^\dagger(y)T^eU(z)U^\dagger(x)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,x,z)\Big[\frac{N_c}{2}\mathrm{Tr}\{U^\dagger(x)U(w')\} - \frac{N_c}{2}\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad - \frac{1}{2}\mathrm{Tr}\{T^eU(w')U^\dagger(x)T^eU(z)U^\dagger(w')\} + \frac{1}{2}\mathrm{Tr}\{T^eU(x)U^\dagger(y)T^eU(y)U^\dagger(z)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,y,z)\Big[\frac{N_c}{2}\mathrm{Tr}\{U^\dagger(w)U(y)\} - \frac{N_c}{2}\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad - \frac{1}{2}\mathrm{Tr}\{T^eU(w)U^\dagger(y)T^eU(z)U^\dagger(w)\} + \frac{1}{2}\mathrm{Tr}\{T^eU(x)U^\dagger(y)T^eU(z)U^\dagger(x)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(x,y,z)\Big[- N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} + N_c\mathrm{Tr}\{U^\dagger(x)U(z)\} \\
&\quad + \frac{1}{2}\mathrm{Tr}\{T^eU(w')U^\dagger(x)T^eU(y)U^\dagger(w')\} - \frac{1}{2}\mathrm{Tr}\{T^eU(w')U^\dagger(x)T^eU(z)U^\dagger(w')\} \\
&\quad + \frac{1}{2}\mathrm{Tr}\{T^eU(w)U^\dagger(x)T^eU(y)U^\dagger(w)\} - \frac{1}{2}\mathrm{Tr}\{T^eU(w)U^\dagger(x)T^eU(z)U^\dagger(w)\}\Big]
\end{aligned} \tag{A.9}
$$

Group I, Row III, N_2.

$$
\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w',z)\Big[- N_c\mathrm{Tr}\{U^\dagger(x)U(w')\} + N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}
$$

$$
\begin{aligned}
&\quad + \mathrm{Tr}\{T^e U(w)U^\dagger(w')T^e U(x)U^\dagger(w)\} - \mathrm{Tr}\{T^e U(w)U^\dagger(x)T^e U(y)U^\dagger(w)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w,z)\Big[- N_c\mathrm{Tr}\{U^\dagger(x)U(w')\} + N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad + \mathrm{Tr}\{T^e U(w)U^\dagger(w')T^e U(w')U^\dagger(x)\} - \mathrm{Tr}\{T^e U(w)U^\dagger(y)T^e U(y)U^\dagger(x)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w',z)\Big[- N_c\mathrm{Tr}\{U^\dagger(w)U(y)\} + N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad - \mathrm{Tr}\{T^e U(w')U^\dagger(x)T^e U(x)U^\dagger(y)\} + \mathrm{Tr}\{T^e U(w)U^\dagger(w')T^e U(y)U^\dagger(w)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w,z)\Big[- N_c\mathrm{Tr}\{U^\dagger(w)U(y)\} + N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad - \mathrm{Tr}\{T^e U(w')U^\dagger(x)T^e U(y)U^\dagger(w')\} + \mathrm{Tr}\{T^e U(w)U^\dagger(w')T^e U(w')U^\dagger(y)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w',z)\Big[- N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad + \mathrm{Tr}\{T^e U(w')U^\dagger(x)T^e U(x)U^\dagger(y)\} - \mathrm{Tr}\{T^e U(w)U^\dagger(w')T^e U(y)U^\dagger(w)\} \\
&\quad + \mathrm{Tr}\{T^e U(w)U^\dagger(x)T^e U(y)U^\dagger(w)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w,z)\Big[- N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad + \mathrm{Tr}\{T^e U(w')U^\dagger(x)T^e U(y)U^\dagger(w')\} - \mathrm{Tr}\{T^e U(w)U^\dagger(w')T^e U(w')U^\dagger(y)\} \\
&\quad + \mathrm{Tr}\{T^e U(w)U^\dagger(x)T^e U(x)U^\dagger(y)\}\Big]
\end{aligned}
\tag{A.10}
$$

Group I, Row IV, N_1.

$$
\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w',z)\Big[- N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad - \mathrm{Tr}\{T^e U(w')U(w)T^e U^\dagger(x)U(z)\} + \mathrm{Tr}\{T^e U(w')U(x)T^e U^\dagger(z)U^\dagger(w')\} \\
&\quad + \mathrm{Tr}\{T^e U(w)U^\dagger(x)T^e U(y)U^\dagger(w)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w,z)\Big[N_c\mathrm{Tr}\{U(w')U(x)\} - N_c\mathrm{Tr}\{U(x)U(y)\} \\
&\quad - \mathrm{Tr}\{T^e U(w)U(w')T^e U^\dagger(x)U(z)\} + \mathrm{Tr}\{T^e U(w)U(y)T^e U^\dagger(x)U(z)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,y,z)\Big[N_c\mathrm{Tr}\{U(x)U(y)\} \\
&\quad + \frac{1}{2}\mathrm{Tr}\{T^e U(w')U(w)T^e U^\dagger(x)U(z)\} - \frac{1}{2}\mathrm{Tr}\{T^e U(w')U(x)T^e U^\dagger(z)U^\dagger(w')\} \\
&\quad + \frac{1}{2}\mathrm{Tr}\{T^e U(w)U(w')T^e U^\dagger(y)U(z)\} - \frac{1}{2}\mathrm{Tr}\{T^e U(w)U(y)T^e U^\dagger(z)U^\dagger(w)\} \\
&\quad - \frac{1}{2}\mathrm{Tr}\{T^e U(x)U(w')T^e U^\dagger(w')U^\dagger(y)\} - \frac{1}{2}\mathrm{Tr}\{T^e U(x)U(w)T^e U^\dagger(w)U^\dagger(y)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w',z)\Big[N_c\mathrm{Tr}\{U^\dagger(w)U(y)\} - N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad - \mathrm{Tr}\{T^e U(w')U(w)T^e U^\dagger(y)U(z)\} + \mathrm{Tr}\{T^e U(w')U(x)T^e U^\dagger(y)U(z)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w,z)\Big[- N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad - \mathrm{Tr}\{T^e U(w)U(w')T^e U^\dagger(y)U(z)\} + \mathrm{Tr}\{T^e U(w)U(y)T^e U^\dagger(z)U^\dagger(w)\} \\
&\quad + \mathrm{Tr}\{T^e U(y)U(w')T^e U^\dagger(w')U(x)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,x,z)\Big[- \frac{N_c}{2}\mathrm{Tr}\{U(w')U(x)\} + \frac{N_c}{2}\mathrm{Tr}\{U(x)U(y)\} \\
&\quad + \frac{1}{2}\mathrm{Tr}\{T^e U(w)U(w')T^e U^\dagger(x)U(z)\} - \frac{1}{2}\mathrm{Tr}\{T^e U(w)U(y)T^e U^\dagger(x)U(z)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,y,z)\Big[- \frac{N_c}{2}\mathrm{Tr}\{U(w)U(y)\} + \frac{N_c}{2}\mathrm{Tr}\{U(x)U(y)\} \\
&\quad + \frac{1}{2}\mathrm{Tr}\{T^e U(w')U(w)T^e U^\dagger(y)U(z)\} - \frac{1}{2}\mathrm{Tr}\{T^e U(w')U(x)T^e U^\dagger(y)U(z)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(x,y,z)\Big[- N_c\mathrm{Tr}\{U(x)U(y)\} \\
&\quad + \frac{1}{2}\mathrm{Tr}\{T^e U(w)U(x)T^e U^\dagger(y)U(z)\}\Big]
\end{aligned}
\tag{A.11}
$$

Group I, Row IV, N_2**.**

$$
\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(x,y,z)\Big[-\frac{1}{2}\mathrm{Tr}\{T^eU(w')U(w)T^eU^\dagger(y)U(z)\}\\
&\qquad+\frac{1}{2}\mathrm{Tr}\{T^eU(w')U(x)T^eU^\dagger(y)U(z)\}-\frac{1}{2}\mathrm{Tr}\{T^eU(w)U(w')T^eU^\dagger(y)U(z)\}\\
&\qquad+\frac{1}{2}\mathrm{Tr}\{T^eU(x)U(w')T^eU^\dagger(w')U^\dagger(y)\}+\frac{1}{2}\mathrm{Tr}\{T^eU(x)U(w)T^eU^\dagger(w)U^\dagger(y)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w',z)\Big[N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad+\mathrm{Tr}\{T^eU(w')U(w)T^eU^\dagger(y)U(z)\}-\mathrm{Tr}\{T^eU(w')U(x)T^eU^\dagger(y)U(z)\}\\
&\qquad-\mathrm{Tr}\{T^eU(w)U^\dagger(x)T^eU(y)U^\dagger(w)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w,z)\Big[N_c\mathrm{Tr}\{U(x)U(y)\}\\
&\qquad+\mathrm{Tr}\{T^eU(w)U(w')T^eU^\dagger(y)U(z)\}-\mathrm{Tr}\{T^eU(w)U(x)T^eU^\dagger(y)U(z)\}\\
&\qquad-\mathrm{Tr}\{T^eU(y)U(w')T^eU^\dagger(w')U(x)\}\Big]
\end{aligned}
\tag{A.12}
$$

Group I, Row V.

$$
\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w',z)\Big[-N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad+\mathrm{Tr}\{T^eU(w')U^\dagger(x)T^eU(x)U^\dagger(z)\}-\mathrm{Tr}\{T^eU(w)U^\dagger(w')T^eU(z)U^\dagger(w)\}\\
&\qquad+\mathrm{Tr}\{T^eU(w)U^\dagger(x)T^eU(y)U^\dagger(w)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w,z)\Big[-N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad+\mathrm{Tr}\{T^eU(w')U^\dagger(x)T^eU(y)U^\dagger(w')\}-\mathrm{Tr}\{T^eU(w)U^\dagger(w')T^eU(w')U^\dagger(z)\}\\
&\qquad+\mathrm{Tr}\{T^eU(w)U^\dagger(y)T^eU(y)U^\dagger(z)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w',z)\Big[N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad-\mathrm{Tr}\{T^eU(w')U^\dagger(x)T^eU(x)U^\dagger(z)\}+\mathrm{Tr}\{T^eU(w)U^\dagger(w')T^eU(z)U^\dagger(w)\}\\
&\qquad-\mathrm{Tr}\{T^eU(w)U^\dagger(x)T^eU(y)U^\dagger(w)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w,z)\Big[N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad-\mathrm{Tr}\{T^eU(w')U^\dagger(x)T^eU(y)U^\dagger(w')\}+\mathrm{Tr}\{T^eU(w)U^\dagger(w')T^eU(w')U^\dagger(z)\}\\
&\qquad-\mathrm{Tr}\{T^eU(w)U^\dagger(x)T^eU(x)U^\dagger(z)\}\Big]
\end{aligned}
\tag{A.13}
$$

Group II, Row I.

$$
\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w,z)\Big[-N_c\mathrm{Tr}\{U(w)U(w')\}-N_c\mathrm{Tr}\{U(x)U(y)\}\\
&\qquad+\mathrm{Tr}\{T^eU(w')U(z)T^eU(x)U^\dagger(w')\}+\mathrm{Tr}\{T^eU(z)U(x)T^eU^\dagger(y)U(w)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w',z)\Big[-N_c\mathrm{Tr}\{U(w)U(w')\}-N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad+\mathrm{Tr}\{T^eU(w')U(x)T^eU^\dagger(y)U^\dagger(z)\}+\mathrm{Tr}\{T^eU(w)U(z)T^eU(y)U^\dagger(w)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,x,z)\Big[\frac{N_c}{2}\mathrm{Tr}\{U(w)U(w')\}+\frac{N_c}{2}\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad-\frac{1}{2}\mathrm{Tr}\{T^eU(w')U^\dagger(x)T^eU(z)U^\dagger(w')\}-\frac{1}{2}\mathrm{Tr}\{T^eU(w)U(y)T^eU^\dagger(x)U^\dagger(z)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,y,z)\Big[\frac{N_c}{2}\mathrm{Tr}\{U(w)U(w')\}+\frac{N_c}{2}\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad-\frac{1}{2}\mathrm{Tr}\{T^eU(w')U(x)T^eU^\dagger(y)U^\dagger(z)\}-\frac{1}{2}\mathrm{Tr}\{T^eU(w)U^\dagger(y)T^eU(z)U^\dagger(w)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(x,y,z)\Big[-N_c\mathrm{Tr}\{U(w)U(w')\}-N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad+\frac{1}{2}\mathrm{Tr}\{T^eU(w')U(x)T^eU^\dagger(y)U^\dagger(z)\}+\frac{1}{2}\mathrm{Tr}\{T^eU(w')U^\dagger(x)T^eU(z)U^\dagger(w')\}\\
&\qquad+\frac{1}{2}\mathrm{Tr}\{T^eU(w)U(x)T^eU^\dagger(y)U^\dagger(z)\}+\frac{1}{2}\mathrm{Tr}\{T^eU(w)U^\dagger(x)T^eU(z)U^\dagger(w)\}\Big]
\end{aligned}
$$

$$
\begin{aligned}
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w',z)\Big[N_c\mathrm{Tr}\{U(w)U(w')\}+N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad-\mathrm{Tr}\{T^eU(w')U(x)T^eU^\dagger(y)U^\dagger(z)\}-\mathrm{Tr}\{T^eU(w)U(z)T^eU(x)U^\dagger(w)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w,z)\Big[N_c\mathrm{Tr}\{U(w)U(w')\}+N_c\mathrm{Tr}\{U(x)U(y)\}\\
&\qquad-\mathrm{Tr}\{T^eU(w')U(z)T^eU(x)U^\dagger(w')\}-\mathrm{Tr}\{T^eU(z)U(y)T^eU^\dagger(x)U(w)\}\Big]
\end{aligned}
\tag{A.14}
$$

Group III, Row I, N_1.

$$
\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w',z)\Big[N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad-\mathrm{Tr}\{T^eU(w')U(x)T^eU^\dagger(z)U^\dagger(w')\}-\mathrm{Tr}\{T^eU(w)U^\dagger(x)T^eU(y)U^\dagger(w)\}\\
&\qquad+\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(z)U(x)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w,z)\Big[-N_c\mathrm{Tr}\{U^\dagger(x)U(w')\}+N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad+\mathrm{Tr}\{T^eU(w)U(w')T^eU^\dagger(x)U^\dagger(z)\}-\mathrm{Tr}\{T^eU(w)U(y)T^eU^\dagger(x)U(z)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,y,z)\Big[-N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad+\frac{1}{2}\mathrm{Tr}\{T^eU(w')U(x)T^eU^\dagger(z)U^\dagger(w')\}+\frac{1}{2}\mathrm{Tr}\{T^eU(w')U^\dagger(x)T^eU(y)U^\dagger(w')\}\\
&\qquad-\frac{1}{2}\mathrm{Tr}\{T^eU(w)U(w')T^eU^\dagger(y)U^\dagger(z)\}+\frac{1}{2}\mathrm{Tr}\{T^eU(w)U(y)T^eU^\dagger(z)U^\dagger(w)\}\\
&\qquad+\frac{1}{2}\mathrm{Tr}\{T^eU(w)U^\dagger(x)T^eU(y)U^\dagger(w)\}-\frac{1}{2}\mathrm{Tr}\{T^eU(z)U(x)T^eU^\dagger(w)U(w')\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w',z)\Big[N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad-\mathrm{Tr}\{T^eU(w')U(x)T^eU^\dagger(y)U(z)\}+\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(z)U(y)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w,z)\Big[N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad-\mathrm{Tr}\{T^eU(w')U^\dagger(x)T^eU(y)U^\dagger(w')\}+\mathrm{Tr}\{T^eU(w)U(w')T^eU^\dagger(y)U^\dagger(z)\}\\
&\qquad-\mathrm{Tr}\{T^eU(w)U(y)T^eU^\dagger(z)U^\dagger(w)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,x,z)\Big[\frac{N_c}{2}\mathrm{Tr}\{U^\dagger(x)U(w')\}-\frac{N_c}{2}\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad-\frac{1}{2}\mathrm{Tr}\{T^eU(w)U(w')T^eU^\dagger(x)U^\dagger(z)\}+\frac{1}{2}\mathrm{Tr}\{T^eU(w)U(y)T^eU^\dagger(x)U(z)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,y,z)\Big[\frac{N_c}{2}\mathrm{Tr}\{U^\dagger(w)U(y)\}-\frac{N_c}{2}\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad+\frac{1}{2}\mathrm{Tr}\{T^eU(w')U(x)T^eU^\dagger(y)U(z)\}-\frac{1}{2}\mathrm{Tr}\{T^eU(z)U(y)T^eU^\dagger(w)U(w')\}\Big]
\end{aligned}
\tag{A.15}
$$

Group III, Row I, N_2.

$$
\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w',z)\Big[-N_c\mathrm{Tr}\{U^\dagger(w)U(y)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(x,y,z)\Big[N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad-\frac{1}{2}\mathrm{Tr}\{T^eU(w')U(x)T^eU^\dagger(y)U(z)\}-\frac{1}{2}\mathrm{Tr}\{T^eU(w')U^\dagger(x)T^eU(y)U^\dagger(w')\}\\
&\qquad+\frac{1}{2}\mathrm{Tr}\{T^eU(w)U(w')T^eU^\dagger(y)U^\dagger(z)\}-\frac{1}{2}\mathrm{Tr}\{T^eU(w)U(x)T^eU^\dagger(y)U(z)\}\\
&\qquad-\frac{1}{2}\mathrm{Tr}\{T^eU(w)U^\dagger(x)T^eU(y)U^\dagger(w)\}+\frac{1}{2}\mathrm{Tr}\{T^eU(z)U(y)T^eU^\dagger(w)U(w')\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w',z)\Big[-N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad+\mathrm{Tr}\{T^eU(w')U(x)T^eU^\dagger(y)U(z)\}+\mathrm{Tr}\{T^eU(w)U^\dagger(x)T^eU(y)U^\dagger(w)\}\\
&\qquad-\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^eU^\dagger(z)U(y)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w,z)\Big[-N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad+\mathrm{Tr}\{T^eU(w')U^\dagger(x)T^eU(y)U^\dagger(w')\}-\mathrm{Tr}\{T^eU(w)U(w')T^eU^\dagger(y)U^\dagger(z)\}
\end{aligned}
$$

$$+ \mathrm{Tr}\{T^e U(w)U(x)T^e U^\dagger(y)U(z)\}\Big] \tag{A.16}$$

Group III, Row II, N_1.

$$\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w',z)\Big[-N_c\mathrm{Tr}\{U(w')U(x)\}-N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad+\mathrm{Tr}\{T^eU(x)U(z)T^eU(y)U^\dagger(x)\}+\mathrm{Tr}\{T^eU(z)U(w')T^eU^\dagger(w')U(x)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w,z)\Big[N_c\mathrm{Tr}\{U^\dagger(w)U(y)\}+N_c\mathrm{Tr}\{U(x)U(y)\}\\
&\qquad-\mathrm{Tr}\{T^eU(x)U^\dagger(y)T^eU(y)U(z)\}-\mathrm{Tr}\{T^eU(z)U(x)T^eU^\dagger(y)U(w)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,y,z)\Big[\frac{N_c}{2}\mathrm{Tr}\{U(w')U(x)\}+\frac{N_c}{2}\mathrm{Tr}\{U(w)U(y)\}\\
&\qquad+N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}-\frac{1}{2}\mathrm{Tr}\{T^eU(w')U(x)T^eU(z)U^\dagger(w')\}\\
&\qquad-\frac{1}{2}\mathrm{Tr}\{T^eU(w)U(y)T^eU(z)U^\dagger(w)\}-\frac{1}{2}\mathrm{Tr}\{T^eU(x)U(z)T^eU(y)U^\dagger(x)\}\\
&\qquad-\frac{1}{2}\mathrm{Tr}\{T^eU(x)U^\dagger(y)T^eU(y)U(z)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w',z)\Big[N_c\mathrm{Tr}\{U(w')U(x)\}+N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad-\mathrm{Tr}\{T^eU(x)U(z)T^eU(y)U^\dagger(x)\}-\mathrm{Tr}\{T^eU(z)U(y)T^eU^\dagger(x)U(w')\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w,z)\Big[-N_c\mathrm{Tr}\{U^\dagger(w)U(y)\}-N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad+\mathrm{Tr}\{T^eU(x)U^\dagger(y)T^eU(y)U(z)\}+\mathrm{Tr}\{T^eU(z)U(w)T^eU^\dagger(w)U(y)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,x,z)\Big[-\frac{N_c}{2}\mathrm{Tr}\{U(w)U(y)\}-\frac{N_c}{2}\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad+\frac{1}{2}\mathrm{Tr}\{T^eU(w)U(y)T^eU^\dagger(x)U^\dagger(z)\}+\frac{1}{2}\mathrm{Tr}\{T^eU(x)U^\dagger(y)T^eU(y)U(z)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,y,z)\Big[-\frac{N_c}{2}\mathrm{Tr}\{U(w')U(x)\}-\frac{N_c}{2}\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad+\frac{1}{2}\mathrm{Tr}\{T^eU(w')U(x)T^eU^\dagger(y)U^\dagger(z)\}+\frac{1}{2}\mathrm{Tr}\{T^eU(x)U(z)T^eU(y)U^\dagger(x)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(x,y,z)\Big[\frac{N_c}{2}\mathrm{Tr}\{U(w)U(x)\}+N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad-\frac{1}{2}\mathrm{Tr}\{T^eU(w)U(x)T^eU^\dagger(y)U^\dagger(z)\}\Big]
\end{aligned}\tag{A.17}$$

Group III, Row II, N_2.

$$\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(x,y,z)\Big[\frac{N_c}{2}\mathrm{Tr}\{U(w')U(x)\}-N_c\mathrm{Tr}\{U(x)U(z)\}\\
&\qquad-\frac{1}{2}\mathrm{Tr}\{T^eU(w')U(x)T^eU^\dagger(y)U^\dagger(z)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w',z)\Big[-N_c\mathrm{Tr}\{U(w')U(x)\}+N_c\mathrm{Tr}\{U(x)U(z)\}\\
&\qquad-N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}+\mathrm{Tr}\{T^eU(z)U(y)T^eU^\dagger(x)U(w')\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w,z)\Big[-N_c\mathrm{Tr}\{U(x)U(y)\}+N_c\mathrm{Tr}\{U(x)U(z)\}\\
&\qquad-N_c\mathrm{Tr}\{U^\dagger(x)U(w)\}+\mathrm{Tr}\{T^eU(z)U(y)T^eU^\dagger(x)U(w)\}\Big]
\end{aligned}\tag{A.18}$$

Group III, Row III.

$$\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w,z)\Big[N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad-\mathrm{Tr}\{T^eU(w')U^\dagger(x)T^eU(x)U(z)\}+\mathrm{Tr}\{T^eU(w)U^\dagger(w')T^eU(z)U^\dagger(w)\}\\
&\qquad-\mathrm{Tr}\{T^eU(w)U^\dagger(x)T^eU(y)U^\dagger(w)\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w',z)\Big[N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}\\
&\qquad-\mathrm{Tr}\{T^eU(w')U^\dagger(x)T^eU(y)U^\dagger(w')\}+\mathrm{Tr}\{T^eU(w)U^\dagger(w')T^eU(w')U^\dagger(z)\}
\end{aligned}$$

$$
\begin{aligned}
&\quad - \mathrm{Tr}\{T^e U(w)U^\dagger(y)T^e U(y)U^\dagger(z)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w',z)\Big[- N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad + \mathrm{Tr}\{T^e U(w')U^\dagger(x)T^e U(y)U^\dagger(w')\} - \mathrm{Tr}\{T^e U(w)U^\dagger(w')T^e U(w')U^\dagger(z)\} \\
&\quad + \mathrm{Tr}\{T^e U(w)U^\dagger(x)T^e U(x)U^\dagger(z)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w,z)\Big[- N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad + \mathrm{Tr}\{T^e U(w')U^\dagger(x)T^e U(x)U(z)\} - \mathrm{Tr}\{T^e U(w)U^\dagger(w')T^e U(z)U^\dagger(w)\} \\
&\quad + \mathrm{Tr}\{T^e U(w)U^\dagger(x)T^e U(y)U^\dagger(w)\}\Big]
\end{aligned}
\tag{A.19}
$$

Group III, Row V.

$$
\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w',z)\Big[N_c\mathrm{Tr}\{U^\dagger(x)U(w')\} - N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad - \mathrm{Tr}\{T^e U(w)U^\dagger(w')T^e U(w')U^\dagger(x)\} + \mathrm{Tr}\{T^e U(w)U^\dagger(y)T^e U(y)U^\dagger(x)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w,z)\Big[N_c\mathrm{Tr}\{U^\dagger(x)U(w')\} - N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad - \mathrm{Tr}\{T^e U(w)U^\dagger(w')T^e U(x)U^\dagger(w)\} + \mathrm{Tr}\{T^e U(w)U^\dagger(x)T^e U(y)U^\dagger(w)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w',z)\Big[N_c\mathrm{Tr}\{U^\dagger(w)U(y)\} - N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad + \mathrm{Tr}\{T^e U(w')U^\dagger(x)T^e U(y)U^\dagger(w')\} - \mathrm{Tr}\{T^e U(w)U^\dagger(w')T^e U(w')U^\dagger(y)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w,z)\Big[N_c\mathrm{Tr}\{U^\dagger(w)U(y)\} - N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad + \mathrm{Tr}\{T^e U(w')U^\dagger(x)T^e U(x)U^\dagger(y)\} - \mathrm{Tr}\{T^e U(w)U^\dagger(w')T^e U(y)U^\dagger(w)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w',z)\Big[N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad - \mathrm{Tr}\{T^e U(w')U^\dagger(x)T^e U(y)U^\dagger(w')\} + \mathrm{Tr}\{T^e U(w)U^\dagger(w')T^e U(w')U^\dagger(y)\} \\
&\quad - \mathrm{Tr}\{T^e U(w)U^\dagger(x)T^e U(x)U^\dagger(y)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w,z)\Big[N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad - \mathrm{Tr}\{T^e U(w')U^\dagger(x)T^e U(x)U^\dagger(y)\} + \mathrm{Tr}\{T^e U(w)U^\dagger(w')T^e U(y)U^\dagger(w)\} \\
&\quad - \mathrm{Tr}\{T^e U(w)U^\dagger(x)T^e U(y)U^\dagger(w)\}\Big]
\end{aligned}
\tag{A.20}
$$

Group III, Row VI.

$$
\begin{aligned}
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w',z)\Big[- N_c\mathrm{Tr}\{U^\dagger(x)U(w')\} + N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad + \mathrm{Tr}\{T^e U(w')U^\dagger(x)T^e U(z)U^\dagger(w')\} - \mathrm{Tr}\{T^e U(x)U^\dagger(y)T^e U(y)U^\dagger(z)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,w,z)\Big[- N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad + \mathrm{Tr}\{T^e U(w)U^\dagger(x)T^e U(y)U^\dagger(w)\} - \mathrm{Tr}\{T^e U(w)U^\dagger(y)T^e U(z)U^\dagger(w)\} \\
&\quad + \mathrm{Tr}\{T^e U(x)U(z)T^e U(y)U^\dagger(x)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w',z)\Big[- N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad + \mathrm{Tr}\{T^e U(w')U^\dagger(x)T^e U(y)U^\dagger(w')\} - \mathrm{Tr}\{T^e U(w')U^\dagger(x)T^e U(z)U^\dagger(w')\} \\
&\quad + \mathrm{Tr}\{T^e U(x)U^\dagger(y)T^e U(y)U^\dagger(z)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,w,z)\Big[- N_c\mathrm{Tr}\{U^\dagger(w)U(y)\} + N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} \\
&\quad + \mathrm{Tr}\{T^e U(w)U^\dagger(y)T^e U(z)U^\dagger(w)\} - \mathrm{Tr}\{T^e U(x)U(z)T^e U(y)U^\dagger(x)\}\Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w',z)\Big[N_c\mathrm{Tr}\{U^\dagger(x)U(y)\} - N_c\mathrm{Tr}\{U^\dagger(x)U(z)\} \\
&\quad - \mathrm{Tr}\{T^e U(w')U^\dagger(x)T^e U(y)U^\dagger(w')\} + \mathrm{Tr}\{T^e U(w')U^\dagger(x)T^e U(z)U^\dagger(w')\}\Big]
\end{aligned}
$$

$$+\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(y,w,z)\Big[-N_c\mathrm{Tr}\{U(x)U(z)\}+N_c\mathrm{Tr}\{U^\dagger(x)U(y)\}$$
$$-\mathrm{Tr}\{T^eU(w)U^\dagger(x)T^eU(y)U^\dagger(w)\}+\mathrm{Tr}\{T^eU(w)U^\dagger(x)T^eU(z)U^\dagger(w)\}\Big] \tag{A.21}$$

B Checks

Scripts in `nlo_gluon_paper/singlet/`:

- `trace.py`: projection and trace form;
- `validate.py`: every term, trace form vs. direct contraction;
- `render_singlet.py`: collection, totals;
- `compare_kernels.py`: Table 1;
- `compare_rows.py`, `compare_rows_fast.py`: Table 2;
- `orient.py`: reading rules for general U;
- `make_singlet.py`: this note.

check	result
trace form vs. direct contraction, your 489 terms (general U)	2.2×10^{-15} (abs.)
the same, JIMWLK (80) and BFKL (7852) target terms	9.5×10^{-16} / 1.2×10^{-15} (abs.)
Eq. (3.1) vs. App. F (corrected), general U	relative difference 1.3×10^{-15}
Eq. (3.2) vs. $-N_c\times$App. E, general U	relative difference 4.2×10^{-16}
rows of Sec. 4, summed, vs. Eq. (3.1), $U=U^T$ / general U	relative difference 1.8×10^{-15} for $U=U^T$, 1.4×10^{-15} for a general U
your lists, summed, vs. Eq. (3.1), $U=U^T$ / general U	4.1×10^{-16} / 3.8×10^{0}
reading rule "every Wilson line acts on its charge", general U	0.13–1.3 (relative); with all lines up to the next f: 0.14–2.8
factored dipole forms expand back to the collected traces	exact (all blocks)